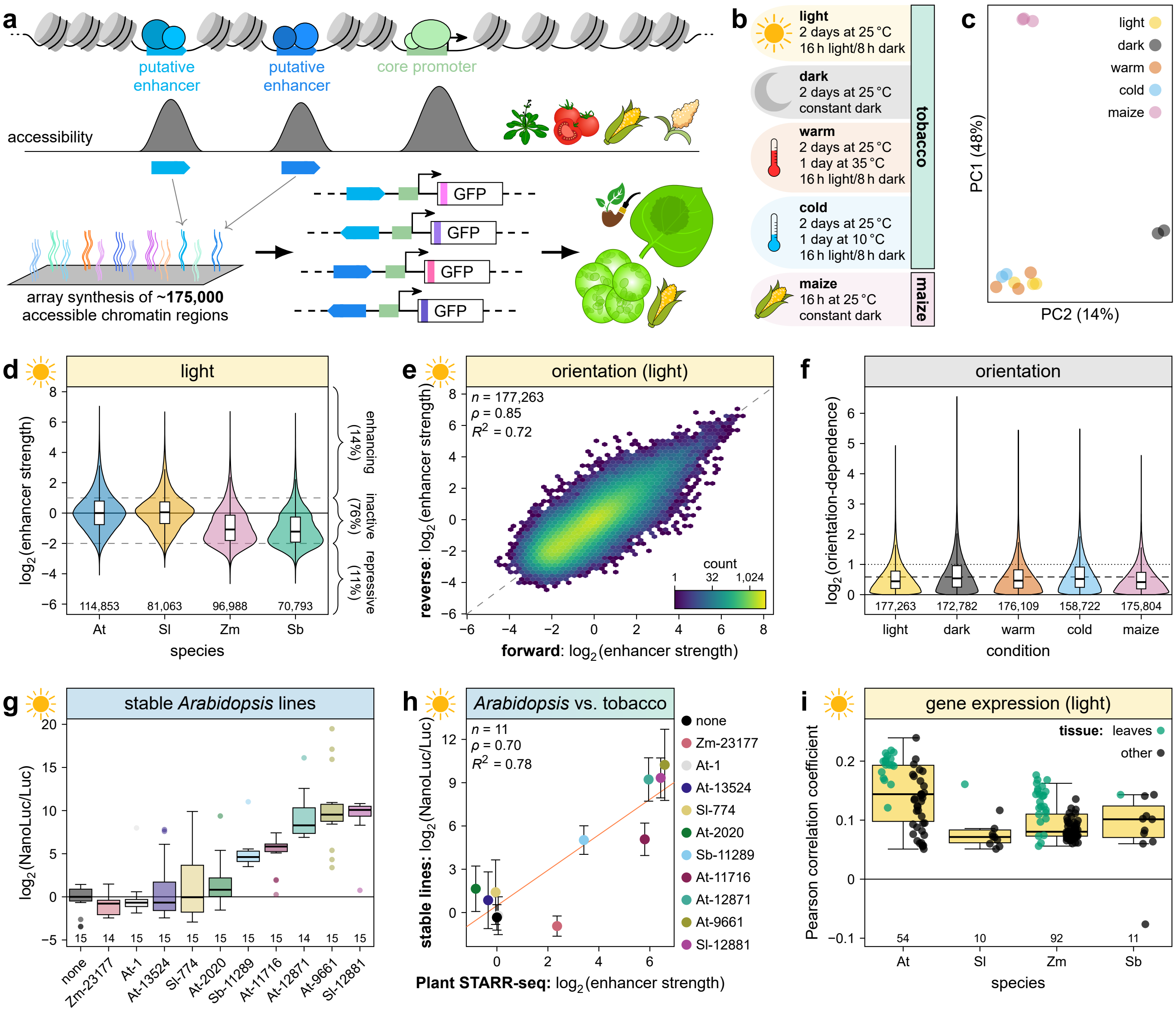


Fig. 1 | Characterization of enhancer strength across species and conditions. a,**b**, Test sequences (170 bp) were derived from accessible chromatin regions (ACRs) in the genomes of *Arabidopsis* (At), tomato (Sl), maize (Zm), and sorghum (Sb). The array-synthesized test sequences were cloned in the forward and reverse orientation upstream of a 35S minimal promoter driving the expression of a barcoded GFP reporter gene. The plasmid library was subjected to Plant STARR-seq in transiently transformed tobacco leaves and maize leaf protoplasts (**a**). After transformation, the plants or protoplasts were subjected to different light and temperature conditions before RNA extraction (**b**). **c**, Principal component analysis of enhancer strength determined in Plant STARR-seq replicate experiments. **d**, Enhancer strength of test sequences grouped by species of origin. The percentage of enhancing ($\log_2$(enhancer strength) > 1), inactive ($\log_2$(enhancer strength) between 1 and −2), and repressive ($\log_2$(enhancer strength) < −2) sequences is indicated. **e**,**f**, Correlation of (**e**) and fold-change in (**f**) enhancer strength between sequences tested in the forward and reverse orientation across the indicated Plant STARR-seq conditions. The dashed and dotted lines in (**e**) represent a 1.5-fold and 2-fold difference, respectively. **g**,**h**, The enhancer strength of ten ACR sequences selected from the Plant STARR-seq library and a negative control without an enhancer (none) was determined using a dual-luciferase assay in stable *Arabidopsis* lines (**g**) and compared to the enhancer strength determined by Plant STARR-seq in tobacco leaves in the light (**h**). **i**, Correlation between enhancer strength and gene expression in various tissues of the indicated species. ACR sequences were linked to the closest transcription start site. For each gene, only the ACR sequence with the highest enhancer strength was retained. Violin plots in (**d**,**f**) represent the kernel density distribution. Box plots inside violin plots and in (**g**,**i**) represent the median (center line), upper and lower quartiles (box limits), 1.5× interquartile range (whiskers), and outliers (points) for all corresponding samples. Numbers at the bottom of each violin or box plot indicate the number of samples in each group. In the hexbin plot in (**d**), color indicates the number of observations in each hexagon. In (**h**), the solid line represents a linear regression line and error bars represent the 95% confidence interval. Pearson's R^2, Spearman's ρ, and the number (n) of samples are indicated in (**e**,**h**).

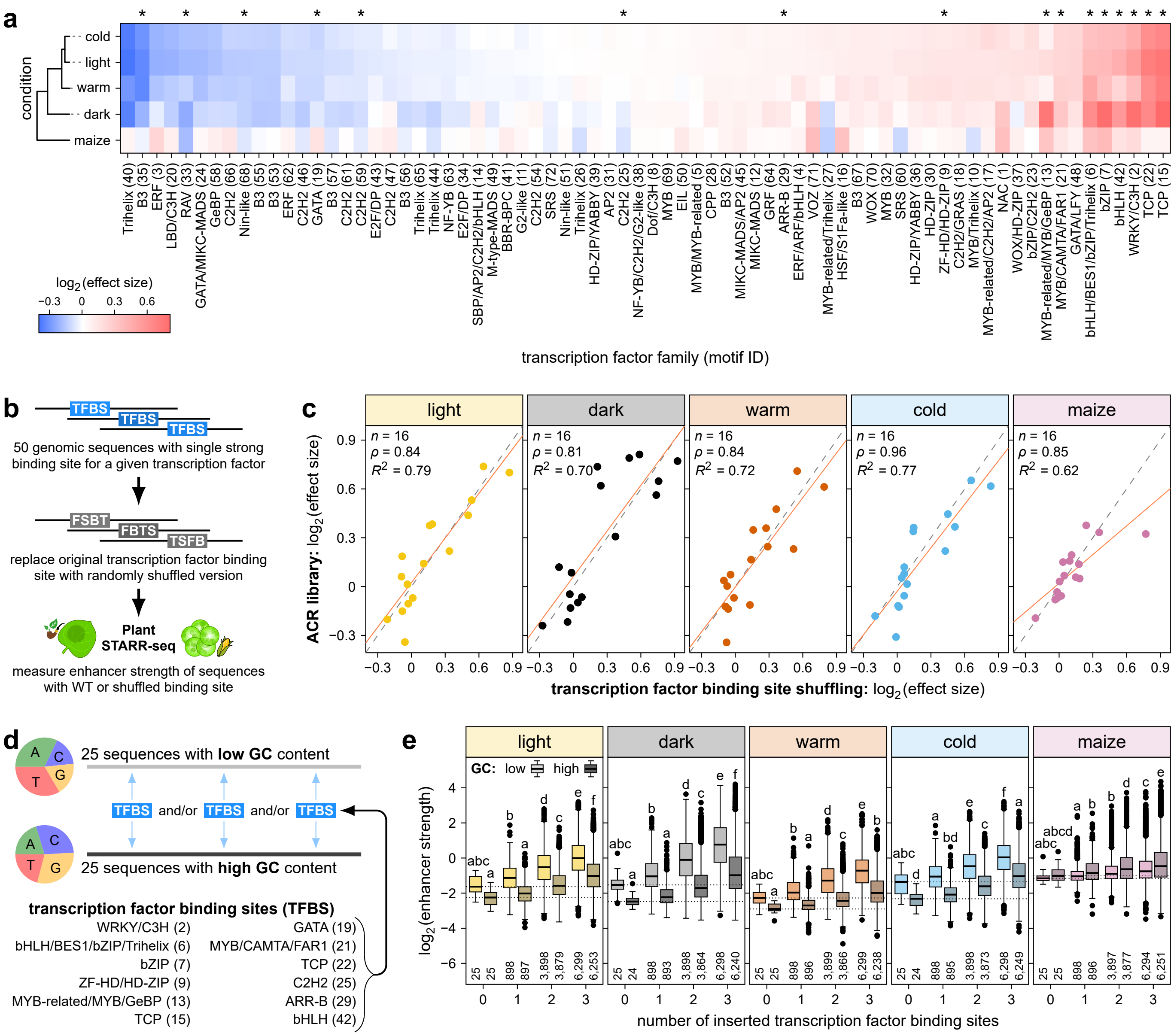


Fig. 2 | Transcription factor binding sites are key determinants of enhancer strength. a, Putative transcription factor binding sites were identified by scanning the test sequences for 72 known binding motifs of the indicated transcription factor families. Effect size, calculated as the fold-change in GC-normalized (see Methods) enhancer strength between test sequences with or without one putative binding site, is indicated by color. **b**,**c**, To test the effect of a transcription factor binding site (TFBS), 50 test sequences containing a corresponding binding site were selected and mutated by replacing the binding site with a shuffled version. All sequences were subjected to Plant STARR-seq in the indicated conditions and species (**b**). The effect size of 16 different binding sites (marked with asterisks in **a**) was determined as the fold-change between the wild-type and mutated sequences, and was compared to the effect size in the ACR library (**c**). The solid and dashed lines represent a linear regression line and a $y = x$ line, respectively. Pearson's R^2, Spearman's ρ, and the number (n) of samples are indicated. **d**,**e**, Synthetic enhancers were designed by inserting one to three transcription factor binding sites into 25 random sequences each with a nucleotide composition similar to an average dicot (low GC) or monocot (high GC) ACR (**d**). The synthetic enhancers were subjected to Plant STARR-seq and their enhancer strengths are shown in box plots as defined in Fig. 1 (**e**). Letters above the box plots indicate significance groups determined by post-hoc Tukey tests performed separately for each condition. Exact p values are listed in Supplementary Data 4.

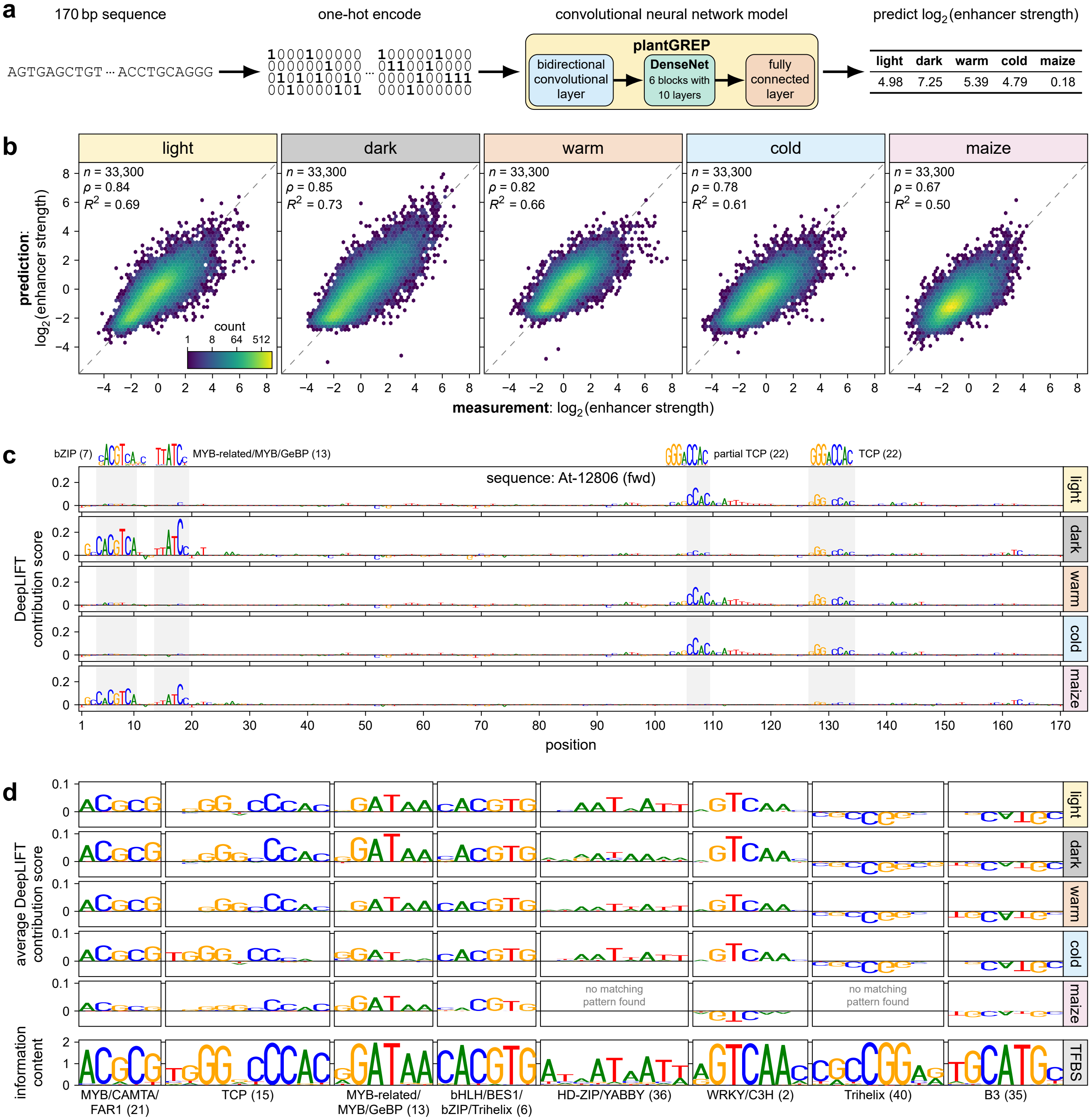


Fig. 3 | A deep learning model predicts enhancer strength and identifies underlying features. a,**b**, A convolutional neural network model, termed plantGREP, consisting of a bidirectional convolutional layer followed by a DenseNet architecture with six blocks of ten layers each and a final fully connected layer was developed to predict enhancer strength across five conditions using one-hot encoded DNA sequences as input (**a**). The model was trained and validated on 90% of the data. The remaining 10% of the data (test set) were used to compare model predictions to the experimentally measured enhancer strength (**b**). The color in the hexbin plots indicate the number of observations in each hexagon and the dashed line represents a $y = x$ line. Pearson's R^2, Spearman's ρ, and the number (n) of samples are indicated. **c**, DeepLIFT was used to calculate contribution scores for an example ACR sequence (At-12806). Known transcription factor binding motifs that match the observed pattern are shown above the plot. **d**, TF-MoDISco was used to identify commonly occurring patterns in the DeepLIFT scores across all sequences of the test set. Examples of patterns detected by TF-MoDISco (top) and matching transcription factor binding motifs (TFBS; bottom) are shown. Patterns were truncated to the size of respective TFBSs.

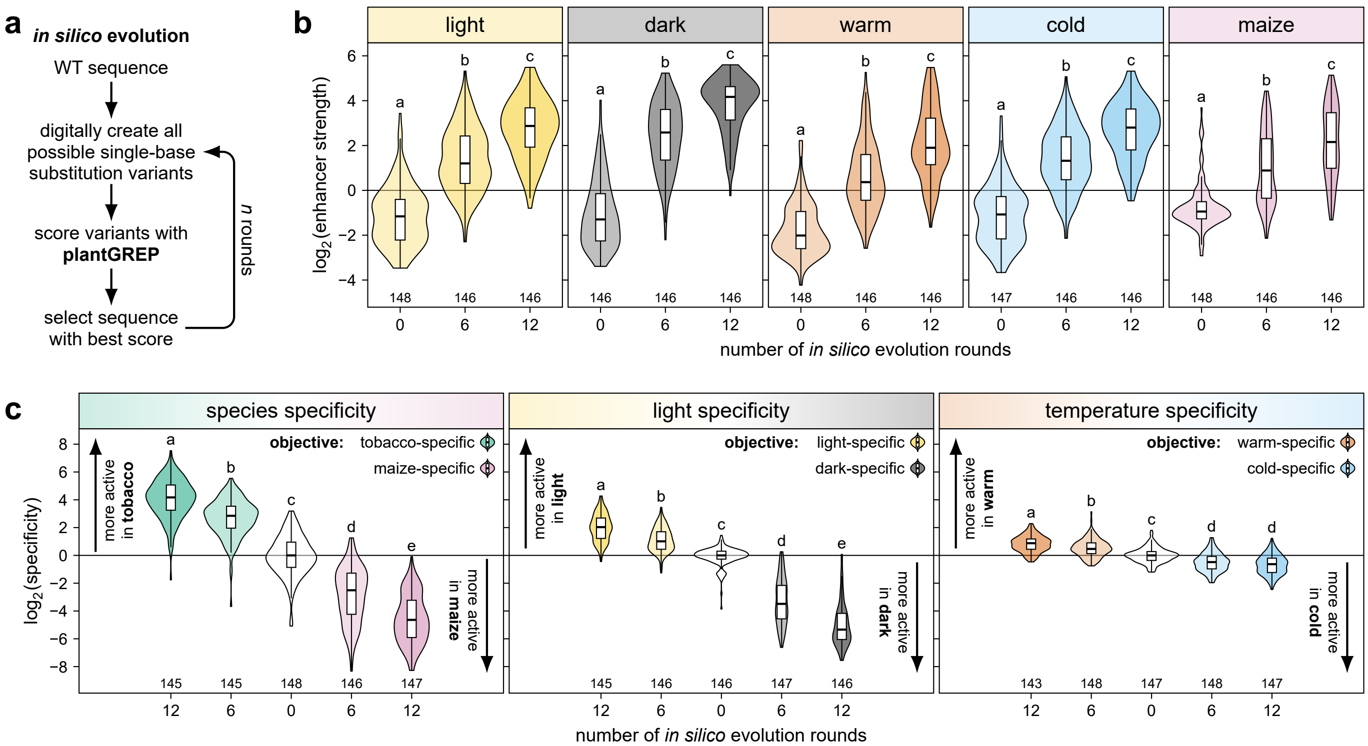


Fig. 4 | Building species- and condition-specific synthetic enhancers through *in silico* evolution. a–**c**, Improved enhancers were generated through *in silico* evolution using plantGREP (**a**). The starting sequences and sequences evolved for six or twelve rounds were subjected to Plant STARR-seq in the indicated conditions and species (**b**,**c**). The objective was to evolve constitutive enhancers (**b**) or species- or condition-specific enhancers (**c**). Species and condition specificity was calculated as the fold-change in enhancer strength between two species (tobacco [mean of light, dark, warm and cold] and maize) or conditions (light and dark for light specificity; warm and cold for temperature specificity). Violin plots in (**b**) and (**c**) are as defined in Fig. 1 and letters above the violin plots indicate significance groups determined by post-hoc Tukey tests performed separately for each condition, species, or specificity. Exact *p* values are listed in Supplementary Data 4.

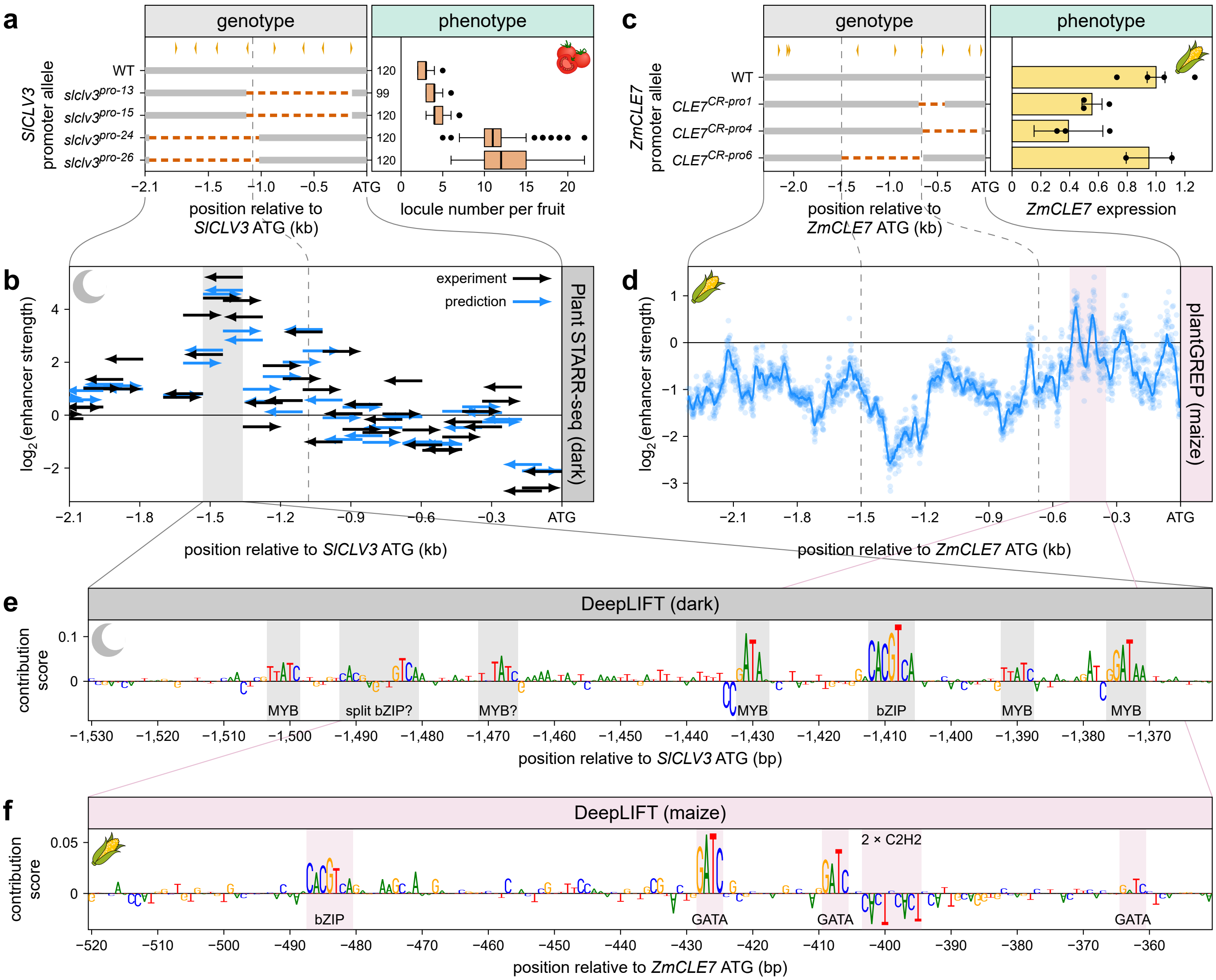
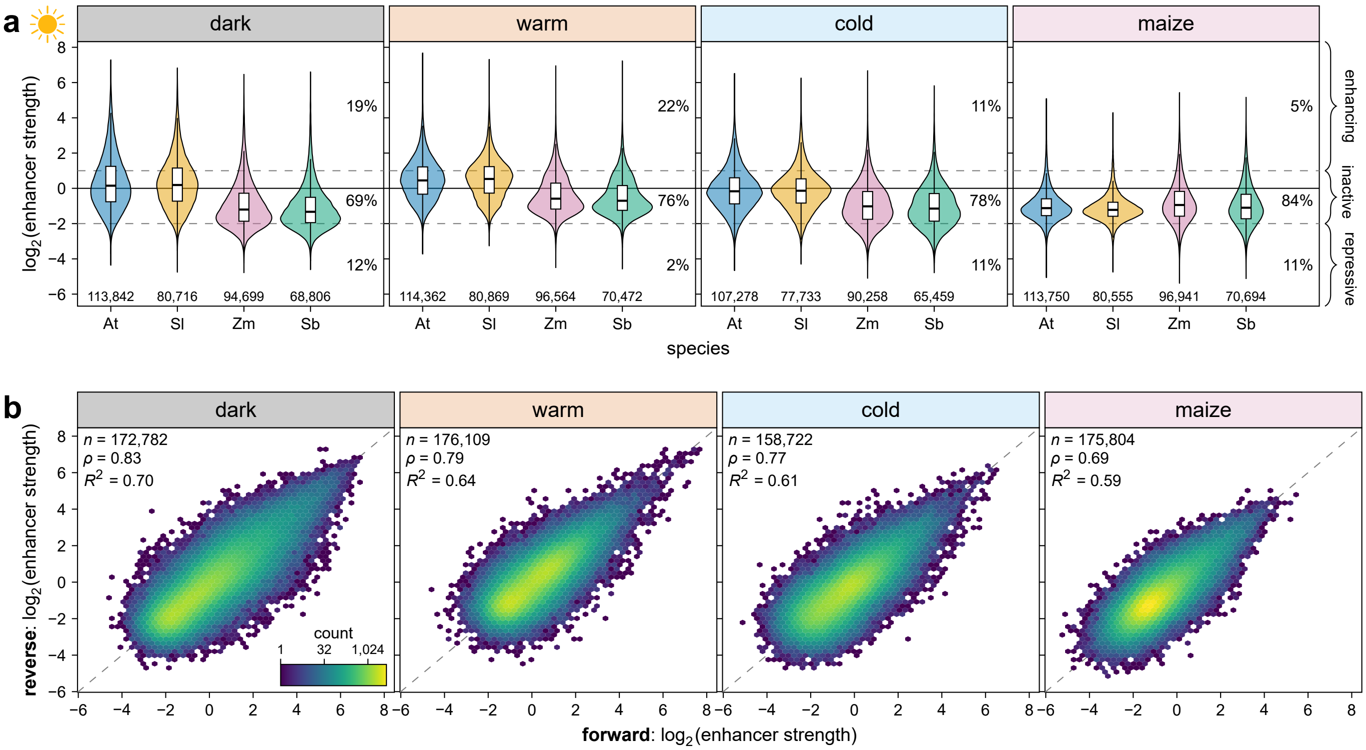
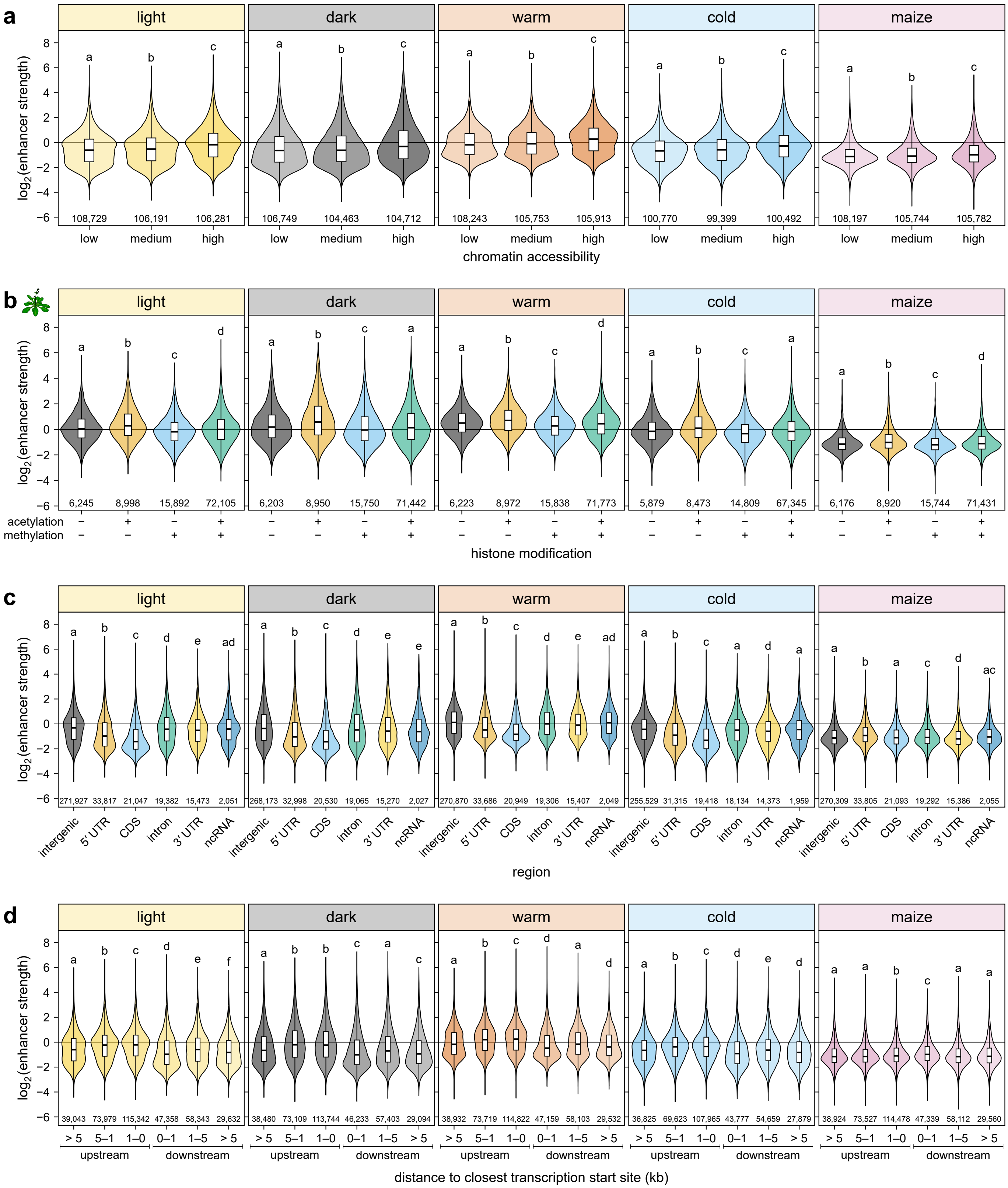


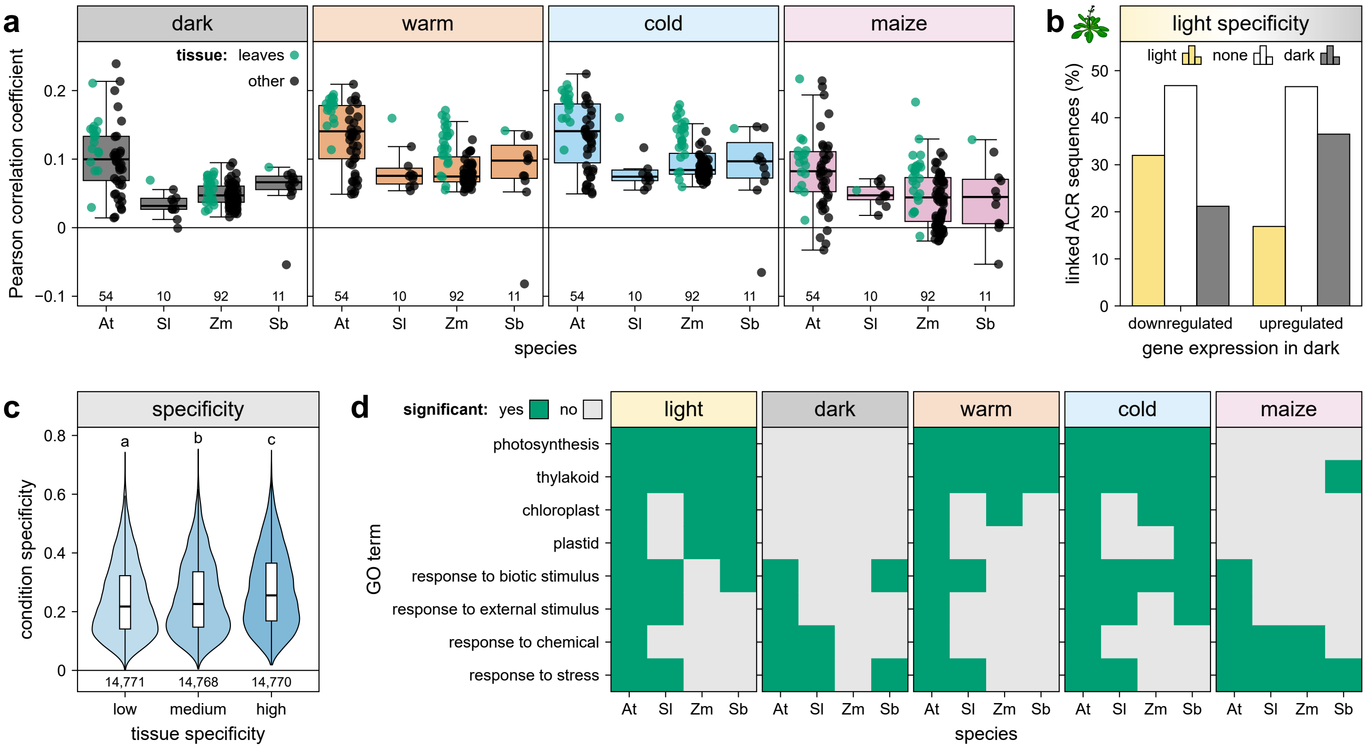
Fig. 5 | Plant STARR-seq data and models can guide genome engineering of regulatory elements. **a**, Wang *et al.* used CRISPR-Cas9 to introduce deletions in the upstream region of *SlCLV3* and measured the number of locules per tomato fruit[18]. Increased numbers of locules are indicative of decreased *SlCLV3* expression levels. **b**, The enhancer strength in the dark of fragments from the *SlCLV3* upstream region was determined experimentally by Plant STARR-seq (black arrows) and predicted using plantGREP (blue arrows). The orientation of the tested fragments is indicated by the direction of the arrows. **c**, Liu *et al.* used CRISPR-Cas9 to introduce deletions in the upstream region of *ZmCLE7* and measured its expression levels[19]. **d**, The enhancer strength in maize of all possible 170-bp fragments from the *ZmCLE7* upstream region was predicted with plantGREP. Points show individual predictions and the line represents a sliding average (window size 20 bp). **e**,**f**, DeepLIFT contribution scores for regions with enhancer activity (shaded in **b** and **d**) of the *SlCLV3* (**e**) and *ZmCLE7* (**f**) upstream regions were calculated with plantGREP for the indicated conditions. Matches to known transcription factor binding sites are shaded and labeled. Data in (**a**) and (**c**) was obtained from ref. [18] and ref. [19], respectively.



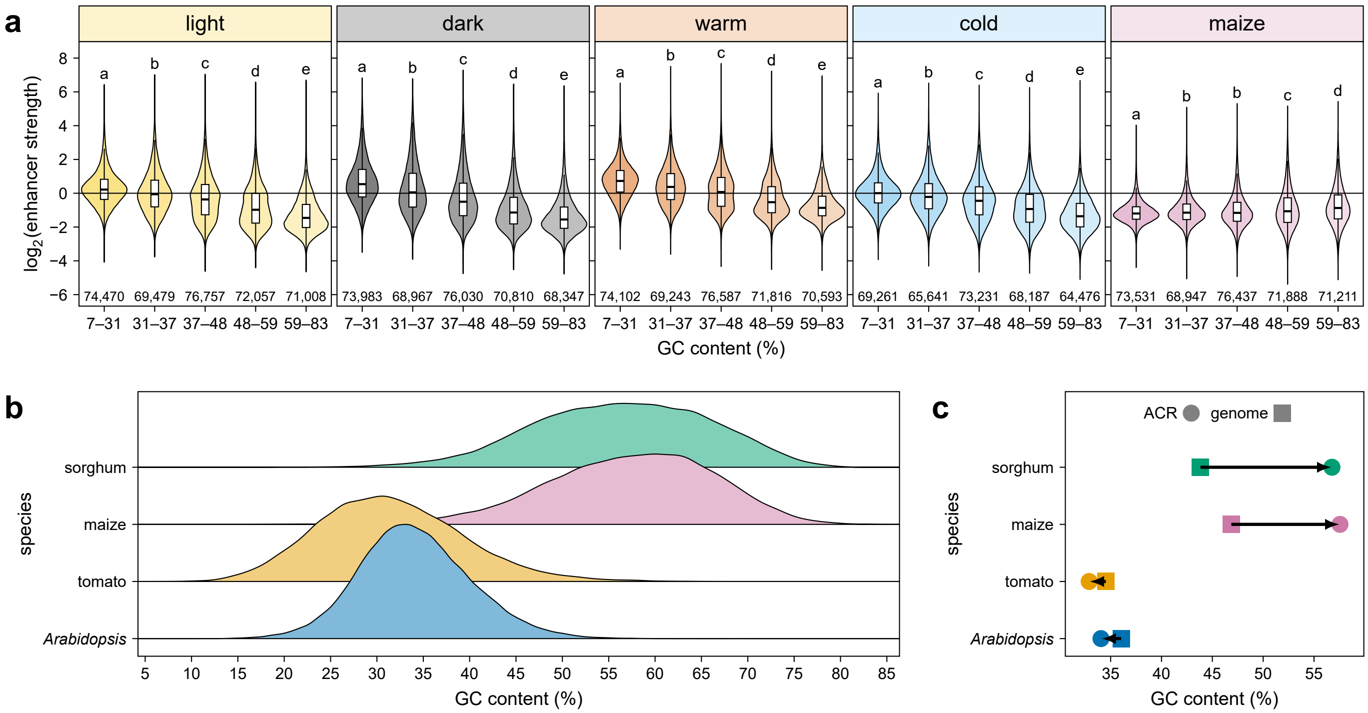
Extended Data Fig. 1 | Regulatory activity spans large dynamic range and is orientation-independent. a, Violin plots (as defined in Fig. 1) of the enhancer strength of test sequences grouped by species of origin. The percentage of enhancing ($\log_2$(enhancer strength) > 1), inactive ($\log_2$(enhancer strength) between 1 and −2), and repressive ($\log_2$(enhancer strength) < −2) sequences is indicated. **b**, Hexbin plots (as defined in Fig. 1) for the correlation of enhancer strength between sequences tested in the forward and reverse orientation.



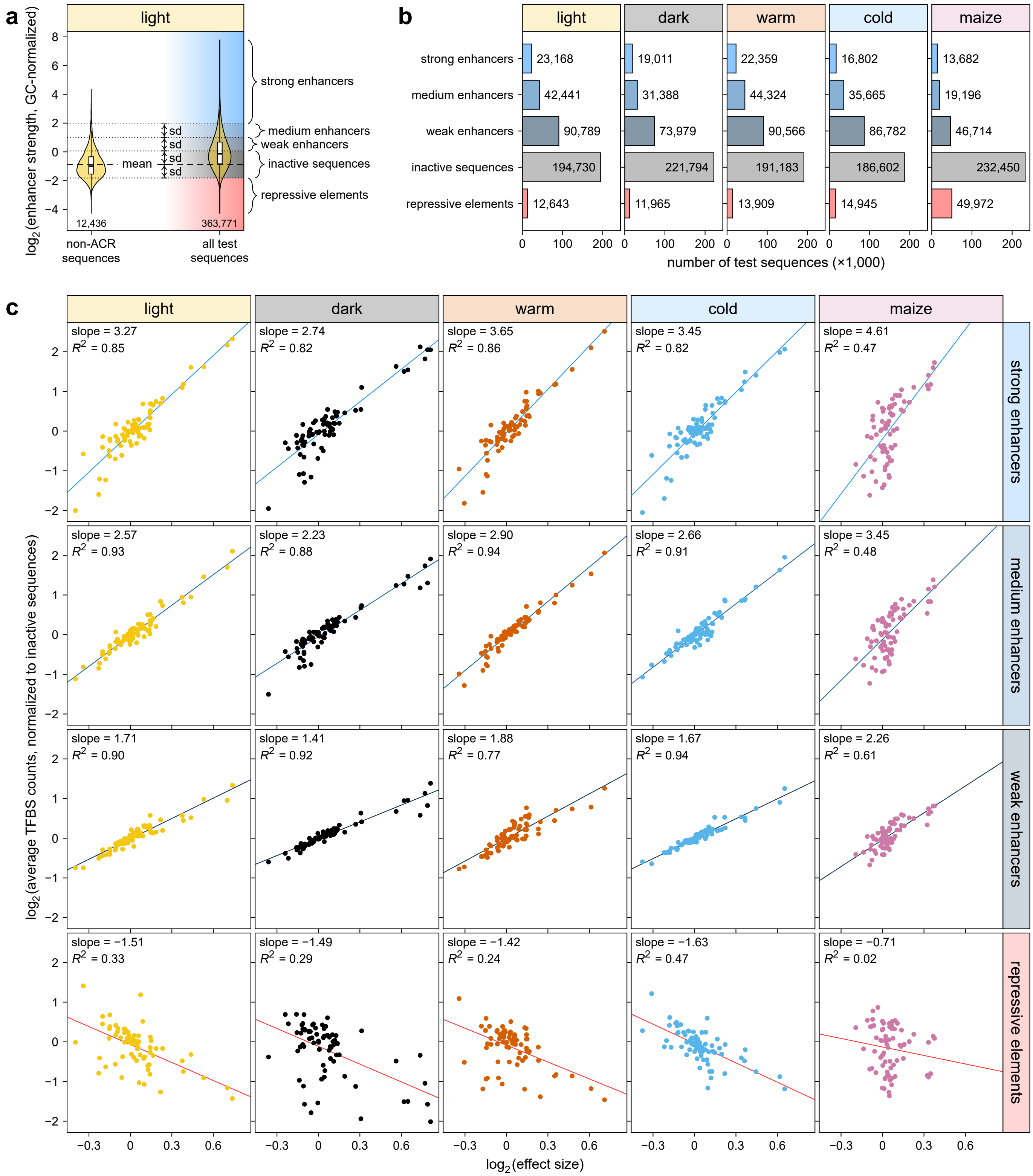
Extended Data Fig. 2 | Genomic signatures of enhancers. a–d, Enhancer strength of test sequences grouped by the relative accessibility of the corresponding region in the genome (**a**), by the occurrence of acetylated and/or methylated histones in a 1-kb window surrounding the corresponding region in the genome (**b**), by the genomic region from which they are derived (**c**), or by their distance to the closest transcription start site (**d**). Violin plots are as defined in Fig. 1 and letters above the plots indicate significance groups determined by post-hoc Tukey tests performed separately for each condition and species. Exact *p* values are listed in Supplementary Data 4. UTR, untranslated region; CDS, coding sequence; ncRNA, non-coding RNA.



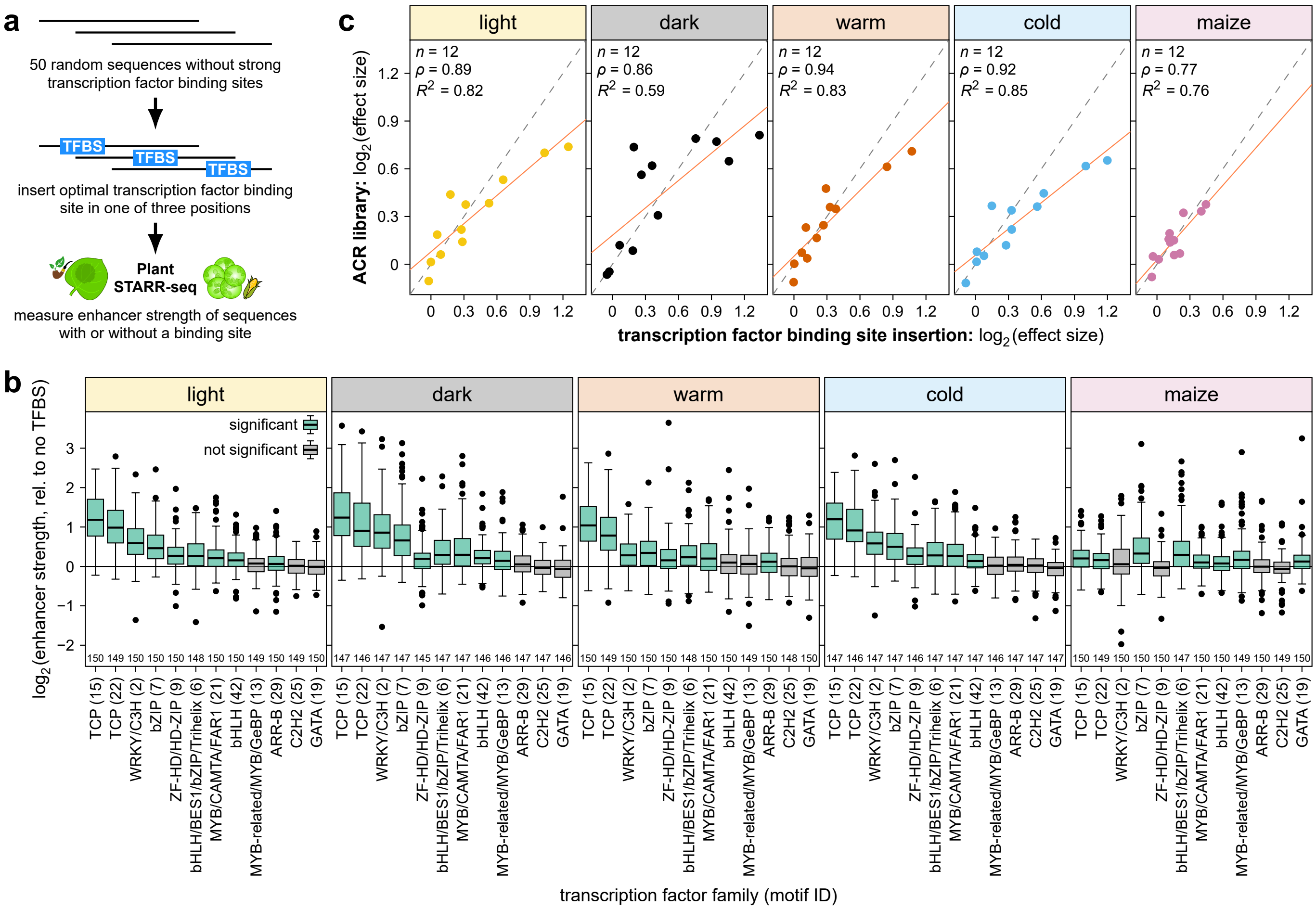
Extended Data Fig. 3 | Enhancer strength is correlated with gene expression *in planta*. a, Correlation between enhancer strength and gene expression in various tissues of the indicated species. At, *Arabidopsis*; Sl, tomato; Zm, maize; Sb, sorghum. ACR sequences were linked to the closest transcription start site. For each gene, only the ACR sequence with the highest enhancer strength was retained. **b**, For genes down- or upregulated after shifting *Arabidopsis* plants to the dark, the percentage of linked ACR sequences showing light-specific (1.5× stronger in the light than in the dark; light), dark-specific (1.5× stronger in the dark than in the light; dark), or constitutive (none) enhancer activity is indicated. **c**, Condition-specific enhancer activity of ACR sequences grouped by the relative tissue specificity of the linked genes. Condition and tissue specificity was determined using the tau index based on our Plant STARR-seq data and the gene expression data used in (**a**), respectively. For each gene, only the ACR sequence with the highest condition specificity was retained. Violin plots are as defined in Fig. 1 and letters above the plots indicate significance groups determined by a post-hoc Tukey test. Exact p values are listed in Supplementary Data 4. **d**, GO term enrichment for genes linked to the top 5% of test sequences with the highest enhancer strength in the indicated conditions and species. Significant (adjusted p value ≤ 0.05) enrichment is indicated in green; non-significant results are in gray. See Supplementary Data 2 for all GO terms and p values. Gene expression data was obtained from ref. [33–36].



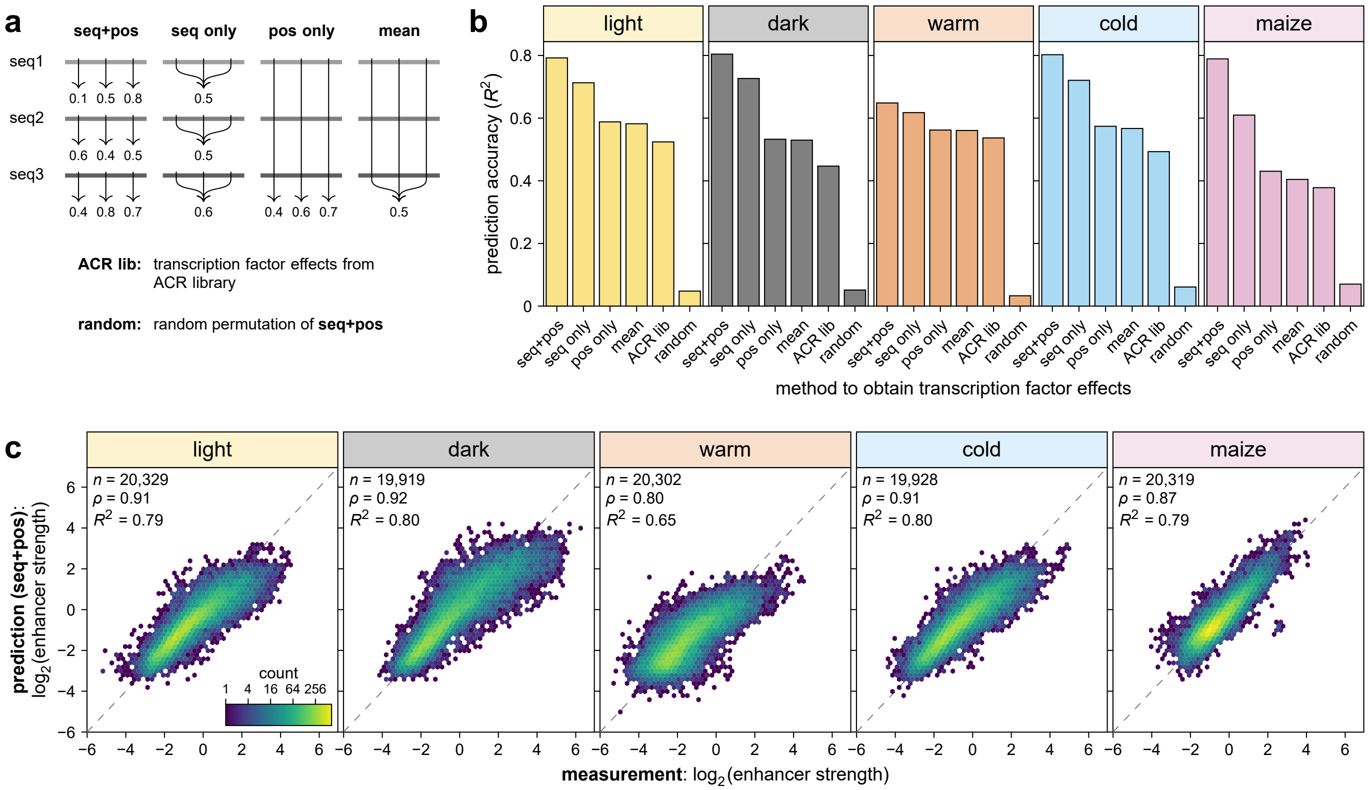
Extended Data Fig. 4 | GC content affects enhancer strength in tobacco. a, Violin plots (as defined in Fig. 1) of the enhancer strength of test sequences grouped by GC content. Letters above the plots indicate significance groups determined by post-hoc Tukey tests performed separately for each condition and species. Exact *p* values are listed in Supplementary Data 4. **b**, Distribution of GC content for the test sequences in the ACR library. **c**, Average GC content of the ACR sequences (circles) and the whole genome (squares) of the indicated species.



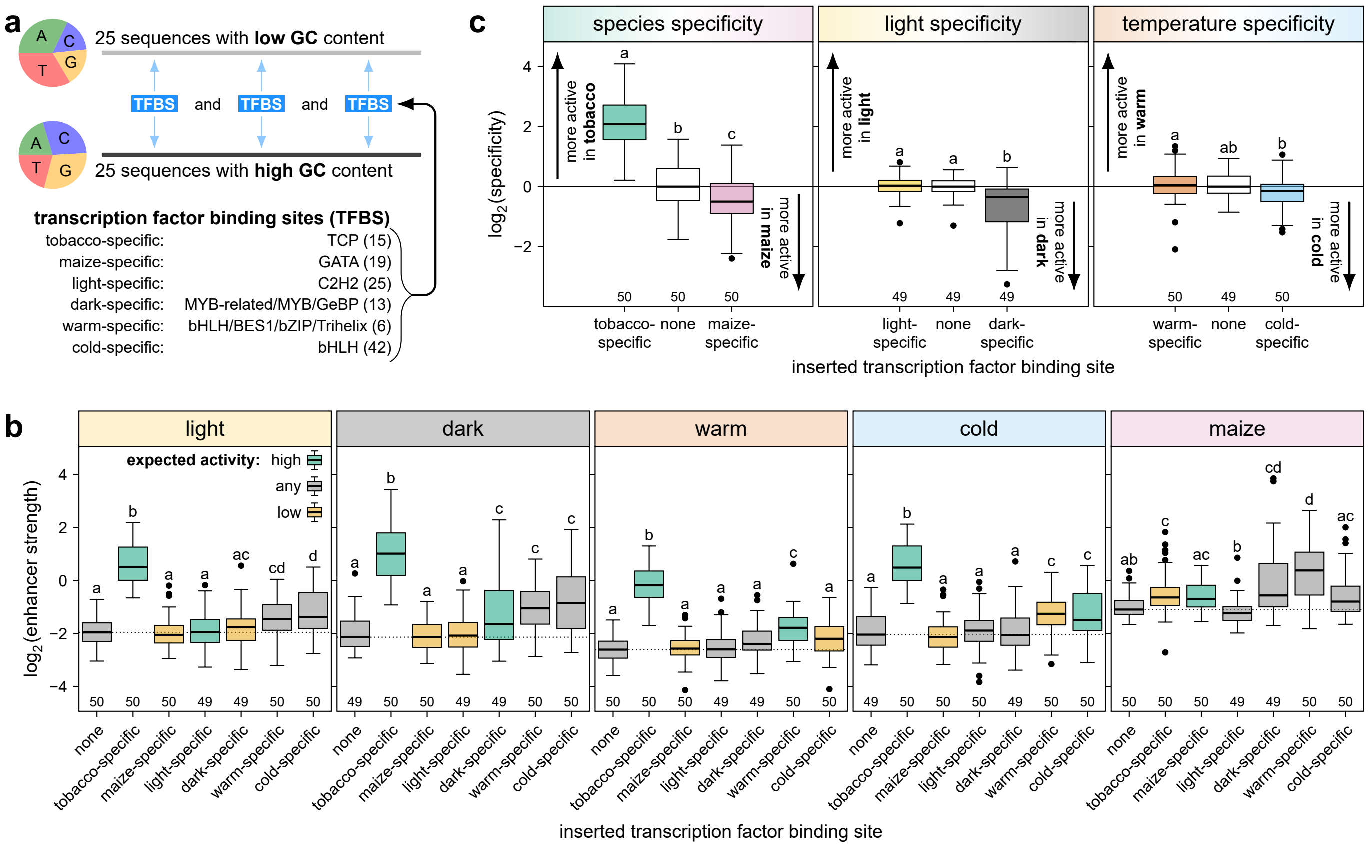
Extended Data Fig. 5 | Activating transcription factor binding sites are enriched in active enhancers. a,**b**, Test sequences in the ACR library were categorized based on the mean (mean) and standard deviation (sd) of the GC-normalized enhancer strength of non-ACR sequences. The categorization scheme is shown for the light condition as an example (**a**). The number of sequences in each category is indicated for all conditions and assay systems (**b**). **c**, The average count of putative transcription factor binding sites (TFBS) in the test sequences of each category was normalized to the average count in inactive sequences and compared to the effect size of the binding site as determined in Fig. 2a.



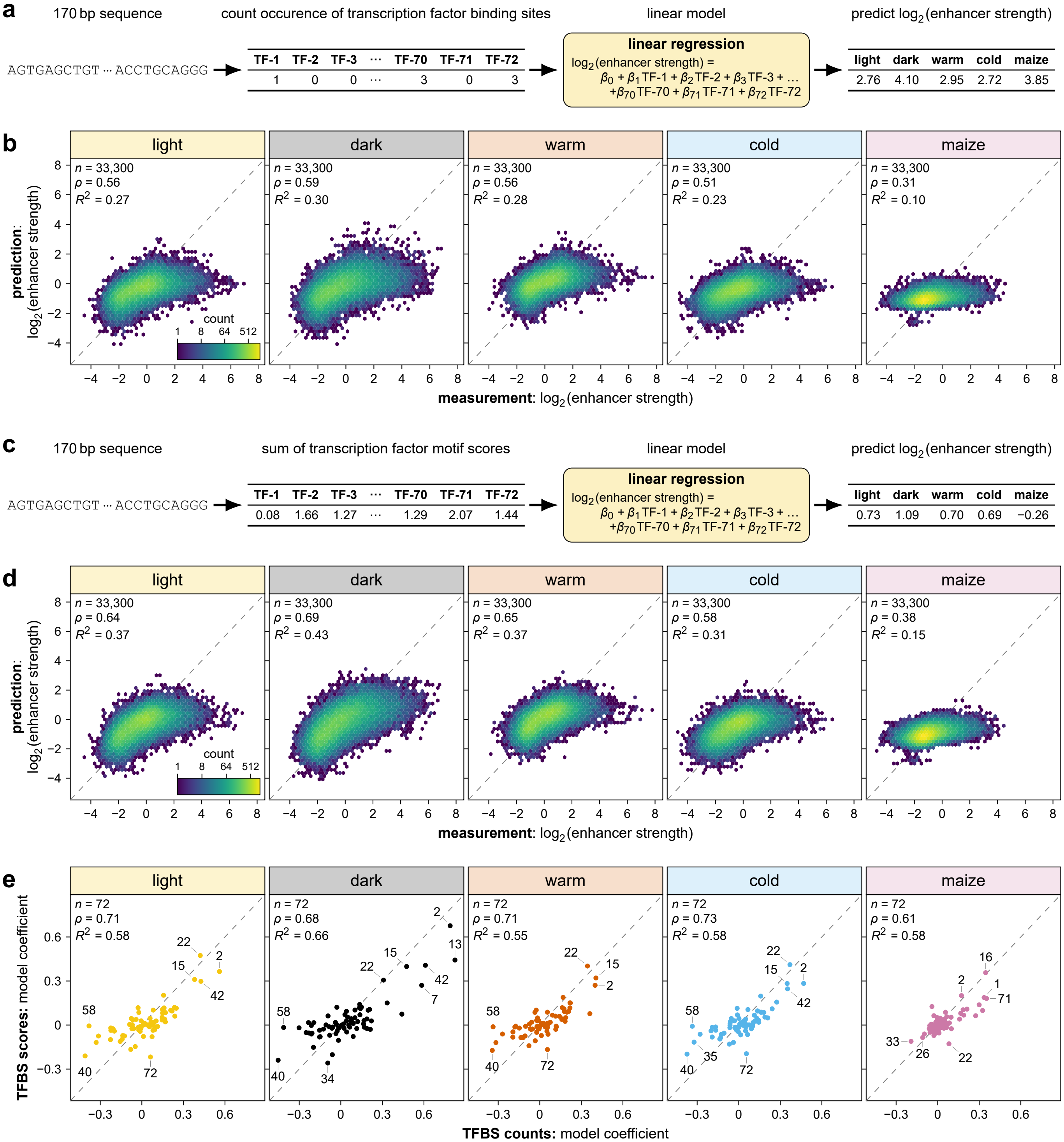
Extended Data Fig. 6 | Inserted transcription factor binding sites can increase enhancer strength. a–**c**, A transcription factor binding site (TFBS) was inserted into 50 random sequences and the sequences were subjected to Plant STARR-seq (**a**) to determine their enhancer strength relative to the corresponding random sequence without a binding site (no TFBS) (**b**). The effect size of each binding site was determined as the fold-change between the sequences with and without it, and was compared to the effect size in the ACR library (see Fig. 2b) (**c**). The solid and dashed lines represent a linear regression line and a $y = x$ line, respectively. Pearson's R^2, Spearman's ρ, and the number (n) of samples are indicated. Box plots in (**b**) are as defined in Fig. 1. Significant differences from a null distribution were determined using one-sample Wilcoxon rank-sum tests. Box plots for significant groups (Bonferroni-adjusted p value ≤ 0.05) are shown in green. Non-significant groups are represented by gray box plots. Exact p values are listed in Supplementary Data 4.



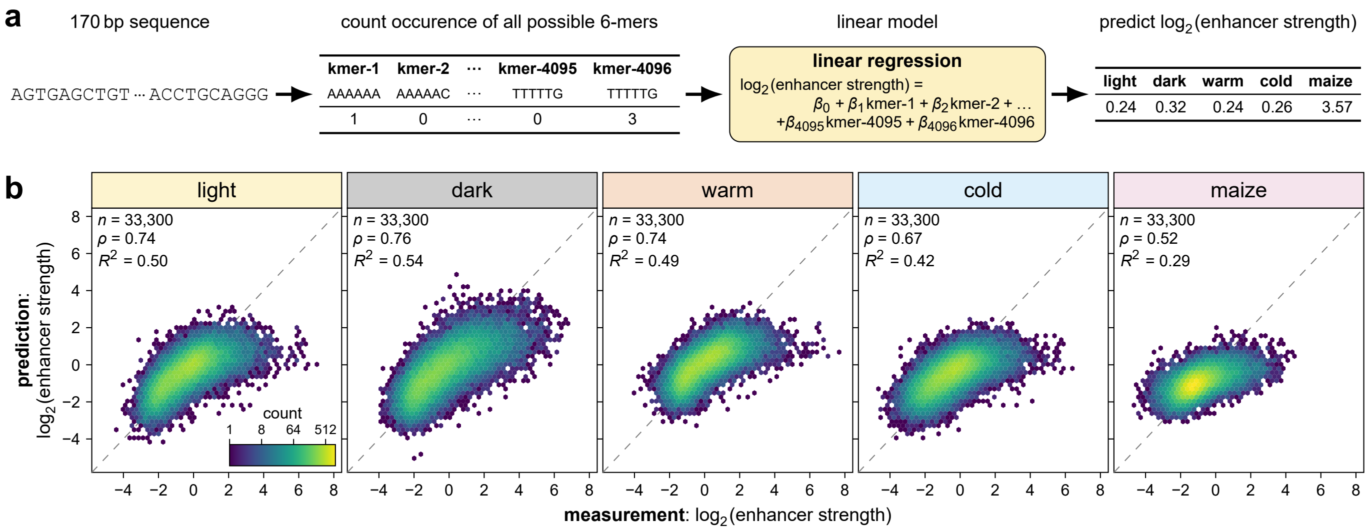
Extended Data Fig. 7 | The strength of synthetic enhancers can be predicted. a–**c**, The strength of synthetic enhancers with two or three transcription factor binding sites was predicted based on the strength of the random sequence without any binding sites and the effects observed for inserting individual binding sites. The transcription factor binding site effects were calculated individually for each random sequence and each position (seq+pos), as the mean across all positions for each sequence (seq only), as the mean across all sequences for each position (pos only), or as the mean across all sequences and positions (mean). Additionally, transcription factor binding site effects were obtained from the ACR library experiment (ACR lib; see Fig. 2a) or from a random permutation of the seq+pos method (**a**). For each method of obtaining transcription factor effects, the prediction accuracy was determined as the correlation between the predicted and experimentally determined synthetic enhancer strengths (**b**). Hexbin plots (as defined in Fig. 1) of this correlation are shown for the seq+pos method (**c**).



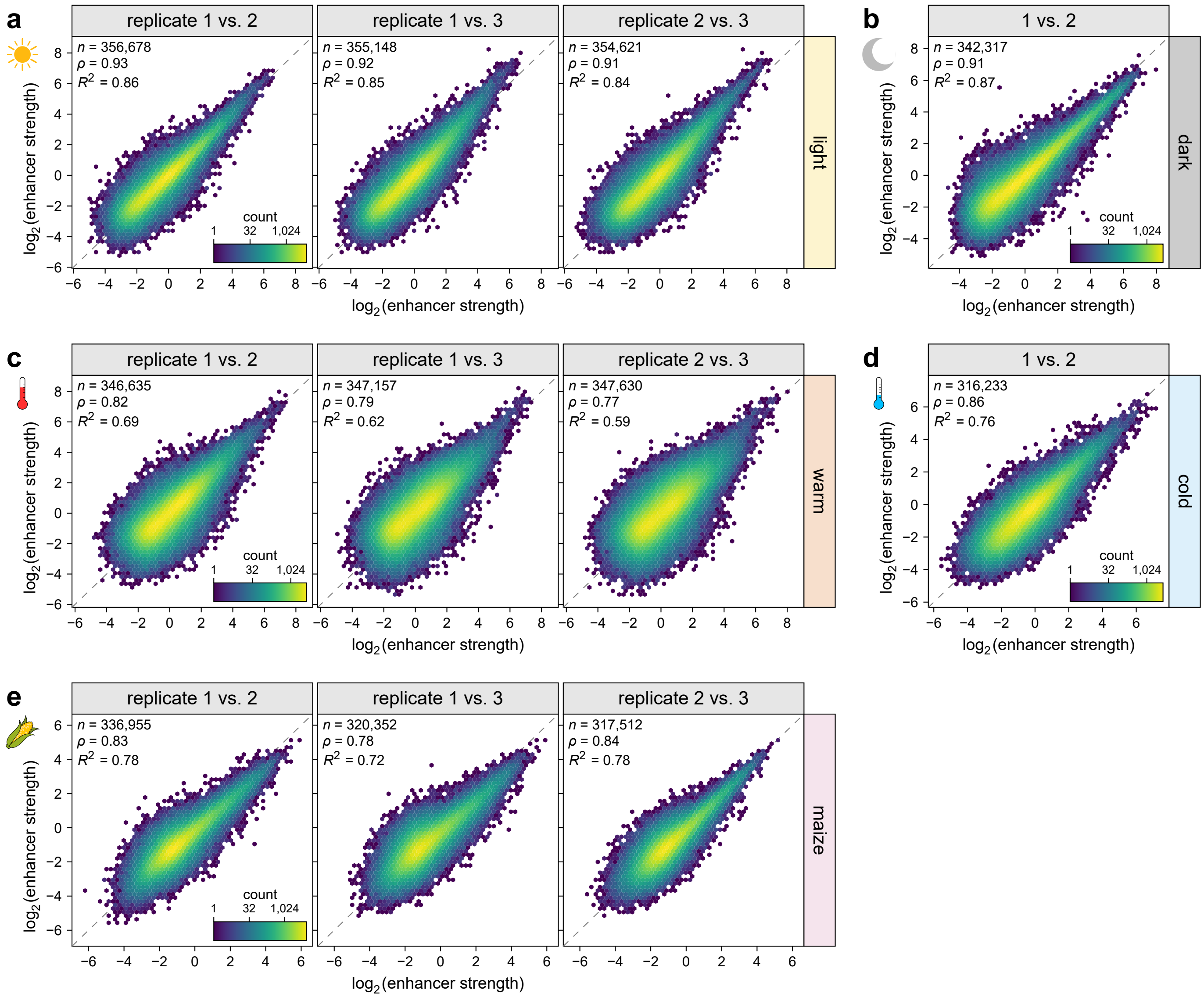
Extended Data Fig. 8 | Rationally designed synthetic enhancers show little condition specificity. a–c, Synthetic enhancers were designed by inserting three binding sites for transcription factors with species- or condition-specific effects into 25 random sequences each with a nucleotide composition similar to an average dicot (low GC) or monocot (high GC) ACR (**a**). The strength (**b**) and species or condition specificity (**c**) of these synthetic enhancers was determined by Plant STARR-seq in the indicated conditions. Species and condition specificity was calculated as the fold-change in enhancer strength between two species (tobacco [mean of light, dark, warm and cold] and maize) or conditions (light and dark for light specificity; warm and cold for temperature specificity). Box plots are as defined in Fig. 1 and letters above the box plots in indicate significance groups determined by post-hoc Tukey tests performed separately for each condition, species, or specificity. Exact *p* values are listed in Supplementary Data 4. In (**b**), box plots are colored according to the expected activity in the indicated condition.



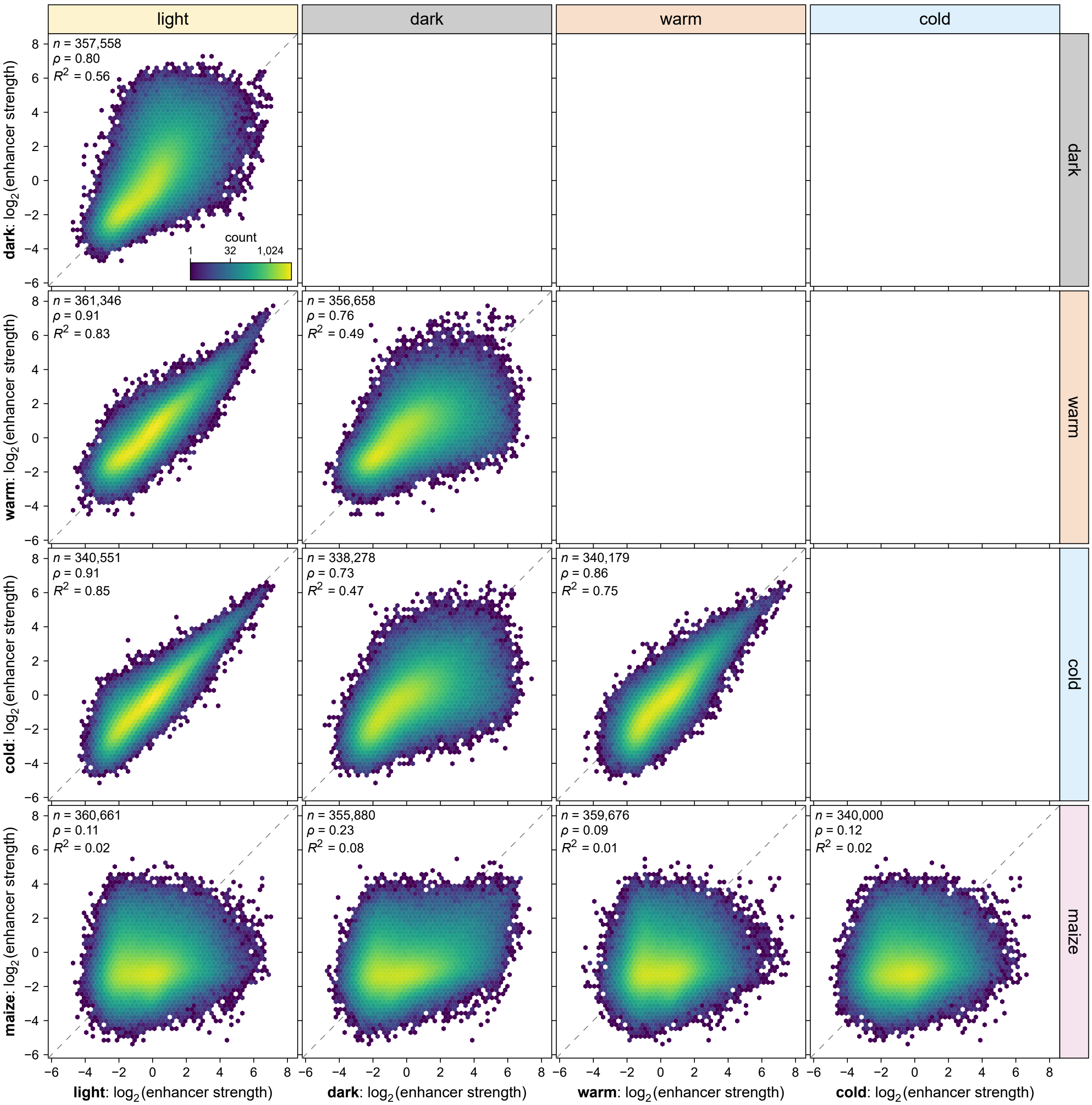
Extended Data Fig. 9 | Linear models based on transcription factor binding sites predict enhancer strength with low accuracy. a–**d**, Linear models based on the counts of putative transcription factor binding sites (**a**,**b**) or on the cumulative scores of binding motif matches within each sequence (**c**,**d**) were trained to predict enhancer strength. The predictions of the models were compared with the experimentally determined enhancer strengths in a held-out test set comprising 10% of the test sequences (**b**,**d**). **e**, Correlation between the coefficients assigned by the linear models in (**a**–**d**) to the 72 different transcription factor binding motifs. Selected motifs associated with increased or decreased enhancer strength are labeled. Hexbin plots in (**b**) and (**d**) are as defined in Fig. 1. In (**b**), (**c**), and (**e**), the dashed lines represents a $y = x$ line. Pearson's R^2, Spearman's ρ, and the number (n) of samples are indicated.



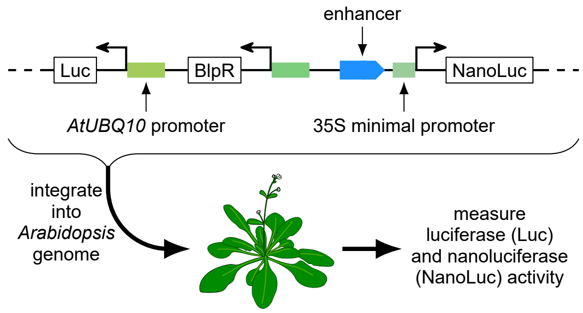
Extended Data Fig. 10 | A linear model based on 6-mer counts predicts enhancer strength with limited accuracy. a,**b**, A linear model based on the counts of all possible 6-mers in a given sequence was trained to predict enhancer strength (**a**). The model predictions were compared with the experimentally determined enhancer strengths in a held-out test set comprising 10% of the test sequences (**b**). Hexbin plots are as defined in Fig. 1.



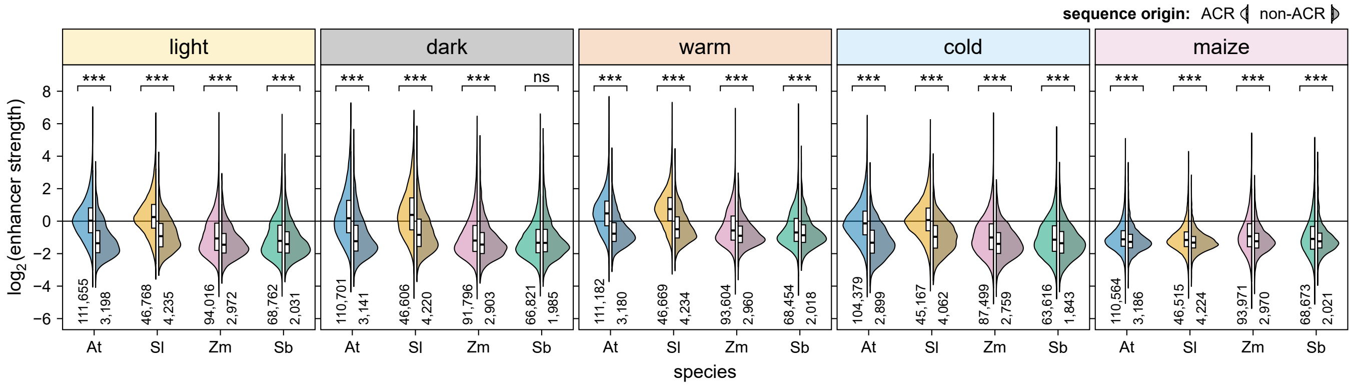
Supplementary Fig. 1 | Plant STARR-seq is reproducible. a–**e**, Hexbin plots (color represents the count of points in each hexagon) of the correlation between enhancer strength in independent Plant STARR-seq replicate experiments in the indicated conditions. The dashed lines represents a $y = x$ line. Pearson's R^2, Spearman's ρ, and the number (n) of samples are indicated.



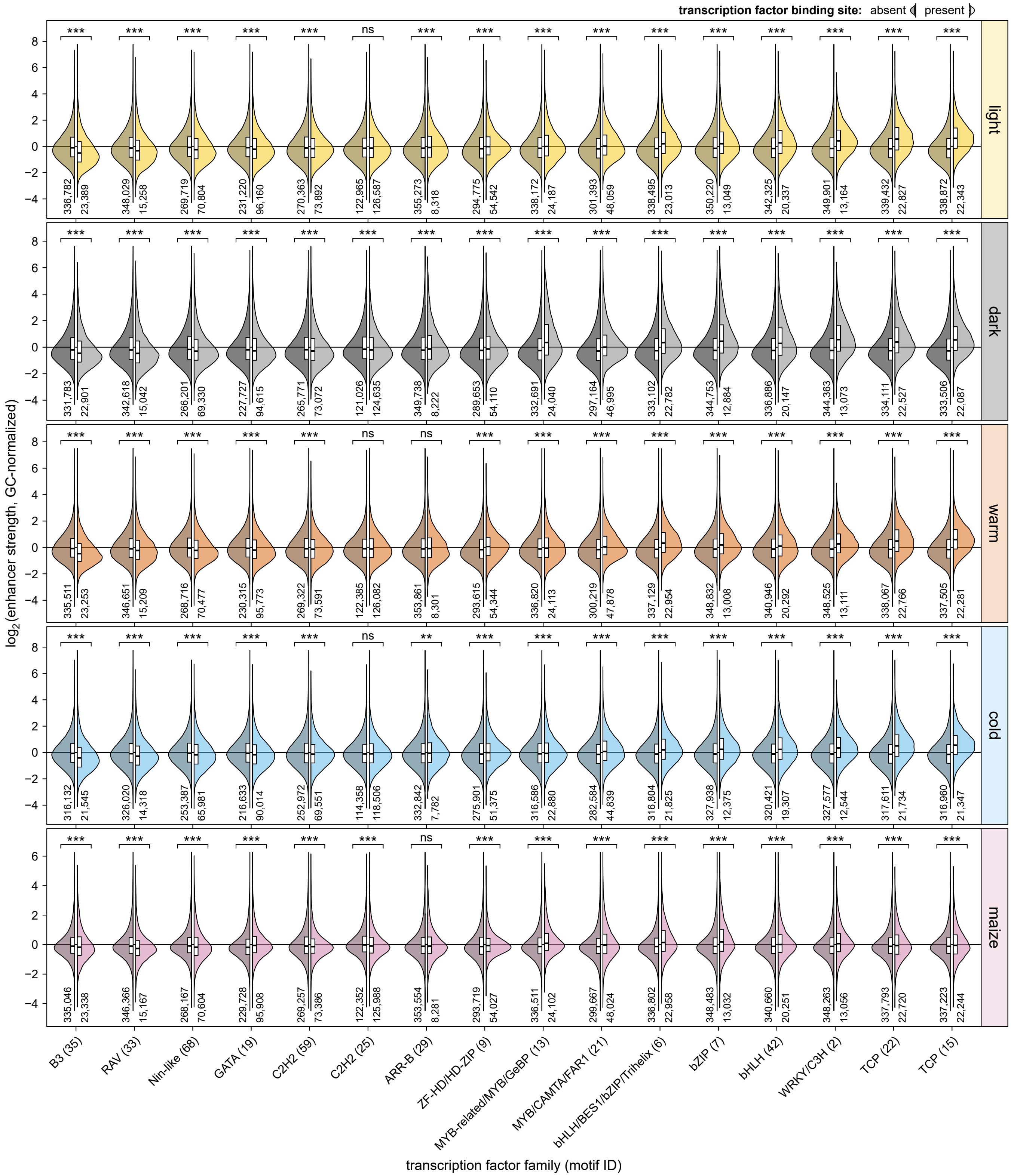
Supplementary Fig. 2 | Plant STARR-seq detects species- and condition-specific differences in enhancer strength. Hexbin plots (as defined in Supplementary Fig. 1) of the correlation between enhancer strength in different conditions.



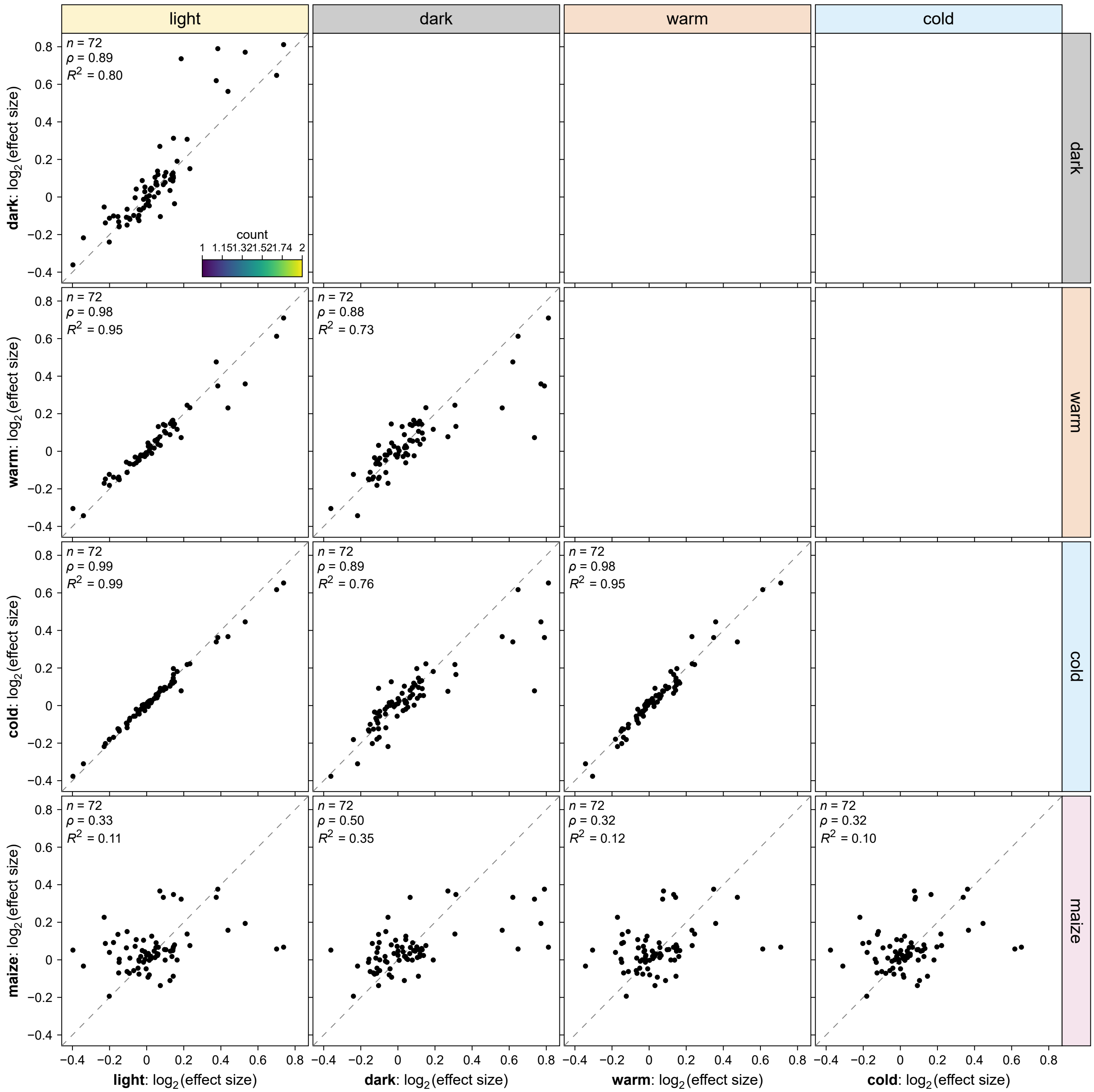
Supplementary Fig. 3 | A dual-luciferase assay to measure enhancer strength in stable *Arabidopsis* lines. Transgenic Arabidopsis lines were generated with T-DNAs harboring a constitutively expressed luciferase (Luc) gene and a nanoluciferase (NanoLuc) gene under control of a 35S minimal promoter coupled to a candidate enhancer. Luciferase and nanoluciferase activity was measured in the T2 generation.



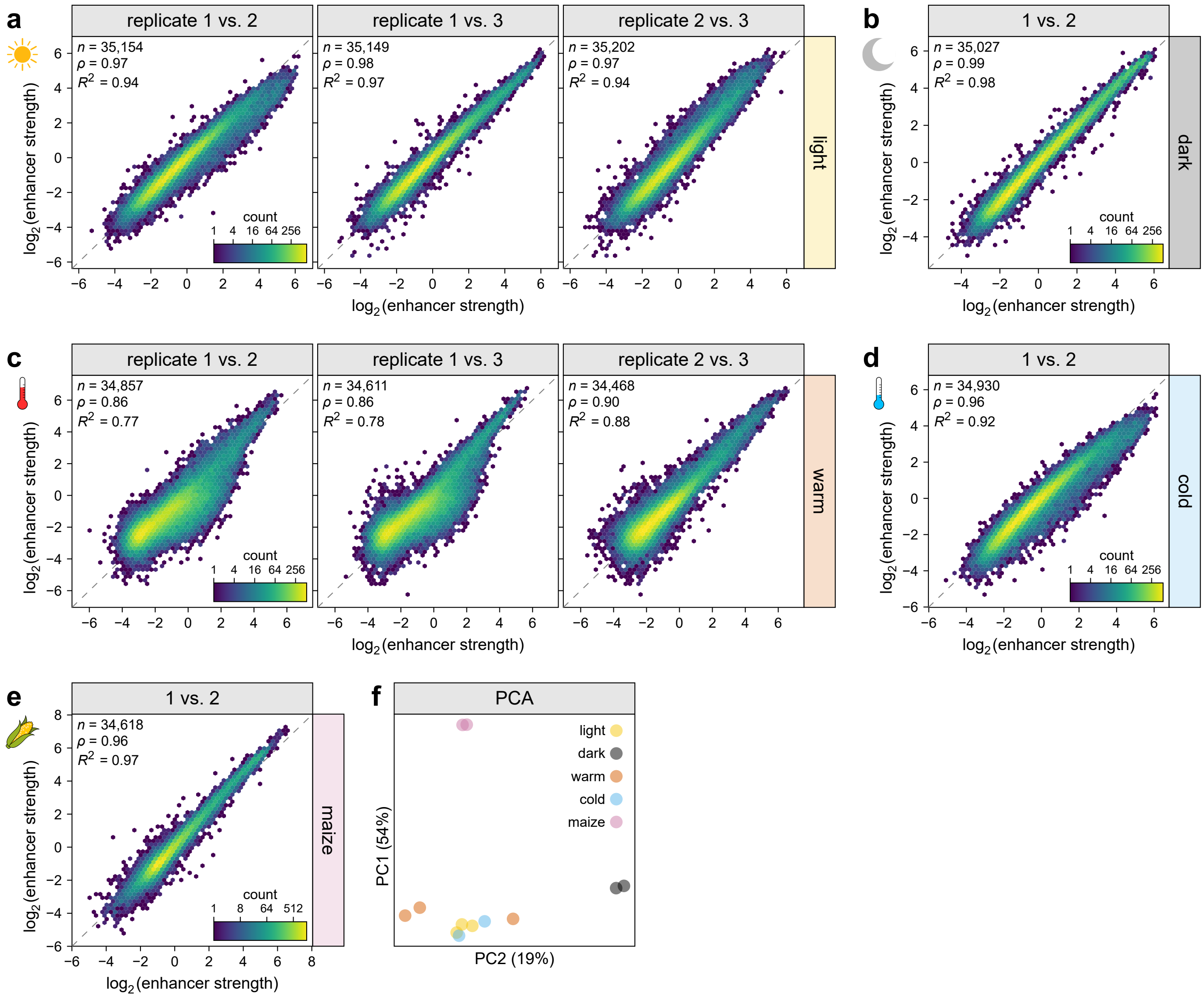
Supplementary Fig. 4 | Accessible chromatin regions are enriched for active enhancers. Enhancer strength of sequences derived from accessible chromatin regions (ACR, left) and non-ACR sequences (right) randomly sampled from inaccessible regions in the genomes of the indicated species. At, *Arabidopsis*; Sl, tomato; Zm, maize; Sb, sorghum. Violin plots represent the kernel density distribution and the box plots inside represent the median (center line), upper and lower quartiles (box limits), and 1.5× interquartile range (whiskers) for all corresponding sequences. Significant differences between ACR and non-ACR sequences were determined by two-sided Wilcoxon rank-sum tests. The obtained p values were Bonferroni-adjusted separately for each condition and species and are indicated: *, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$; ns, not significant. Exact p values are listed in Supplementary Data 4.



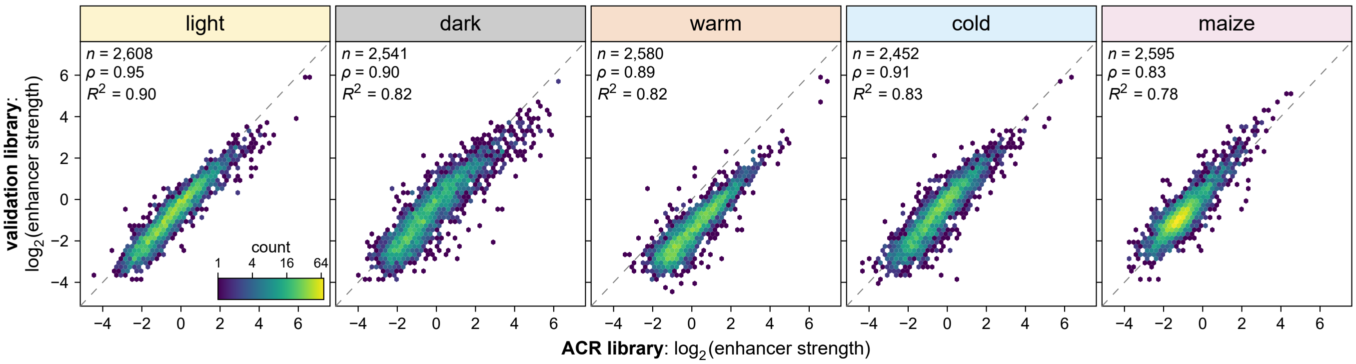
Supplementary Fig. 5 | The presence of transcription factor binding sites affects enhancer strength. Putative transcription factor binding sites were identified by scanning the test sequences with known binding motifs for transcription factors of the indicated families. The enhancer strength of test sequences without or with one putative binding site is shown in violin plots as defined in Supplementary Fig. 4. Significant differences between sequences with and without a binding site were determined by two-sided Wilcoxon rank-sum tests. The obtained p values were Bonferroni-adjusted separately for each condition and species and are indicated: *, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$; ns, not significant. Exact p values are listed in Supplementary Data 4.



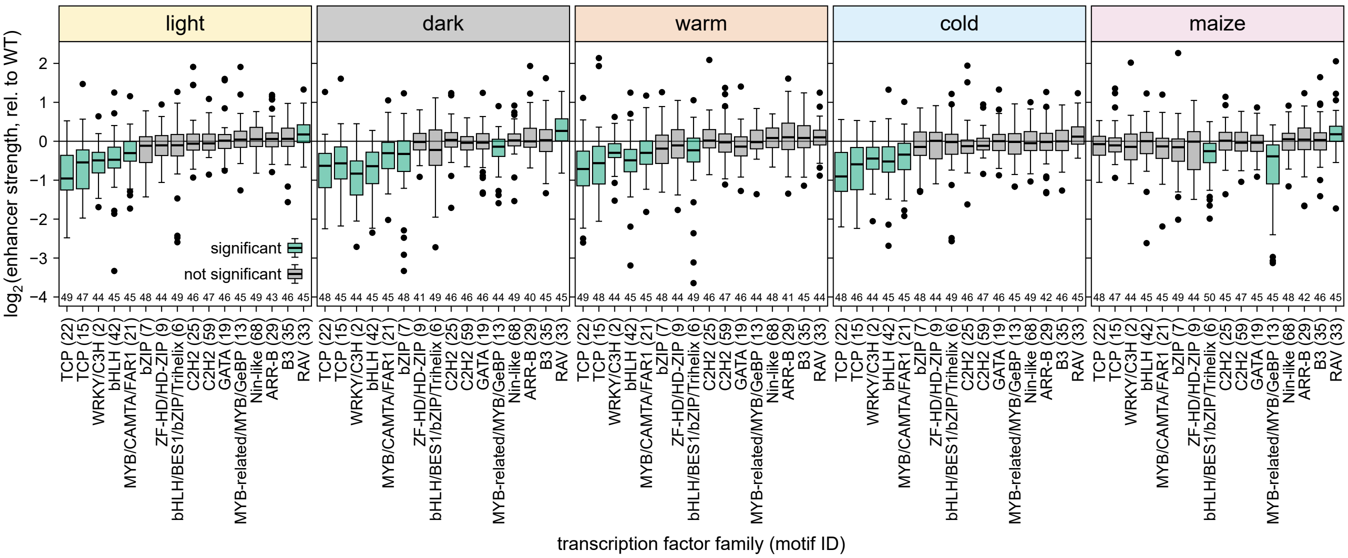
Supplementary Fig. 6 | Transcription factor binding sites show species- and condition-specific effects on enhancer strength. Correlation between the effects of transcription factor binding sites (as determined in Fig. 2) in different conditions. Each dot represents a different transcription factor binding motif. Pearson's R^2, Spearman's ρ, and the number (n) of samples are indicated.



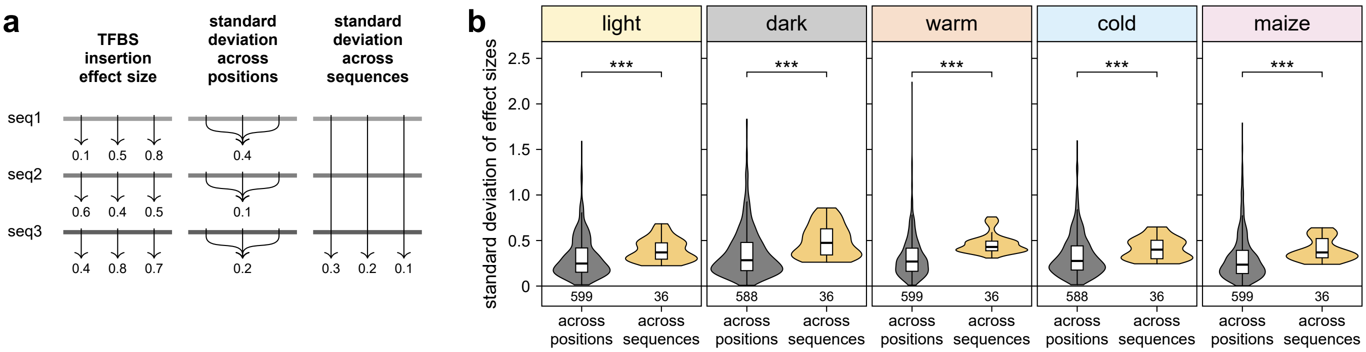
Supplementary Fig. 7 | The validation library yielded reproducible results. a–**e**, Hexbin plots (as defined in Supplementary Fig. 1) of the correlation between enhancer strength in independent Plant STARR-seq replicate experiments with the validation library in the indicated conditions. **f**, Principal component analysis of enhancer strength determined in Plant STARR-seq replicate experiments with the validation library.



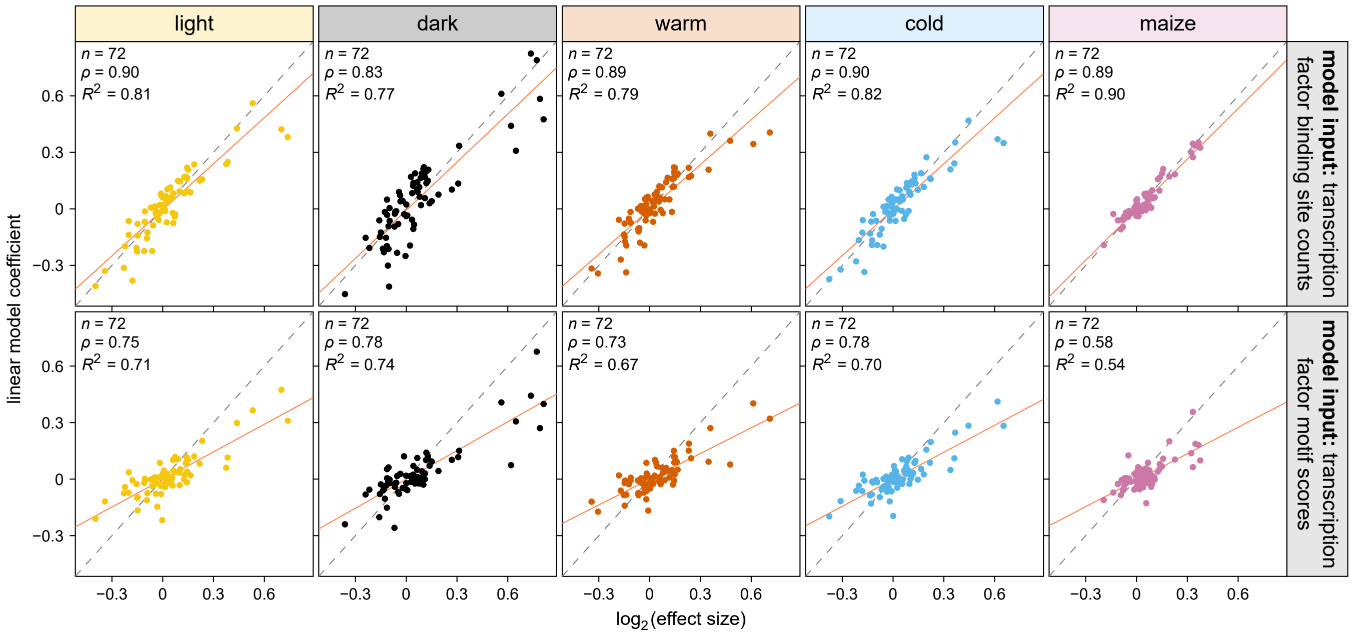
Supplementary Fig. 8 | The validation library results reproduce the results obtained with the ACR library. Hexbin plots (as defined in Supplementary Fig. 1) of the correlation between the strength of enhancers included in the ACR library and the validation library.



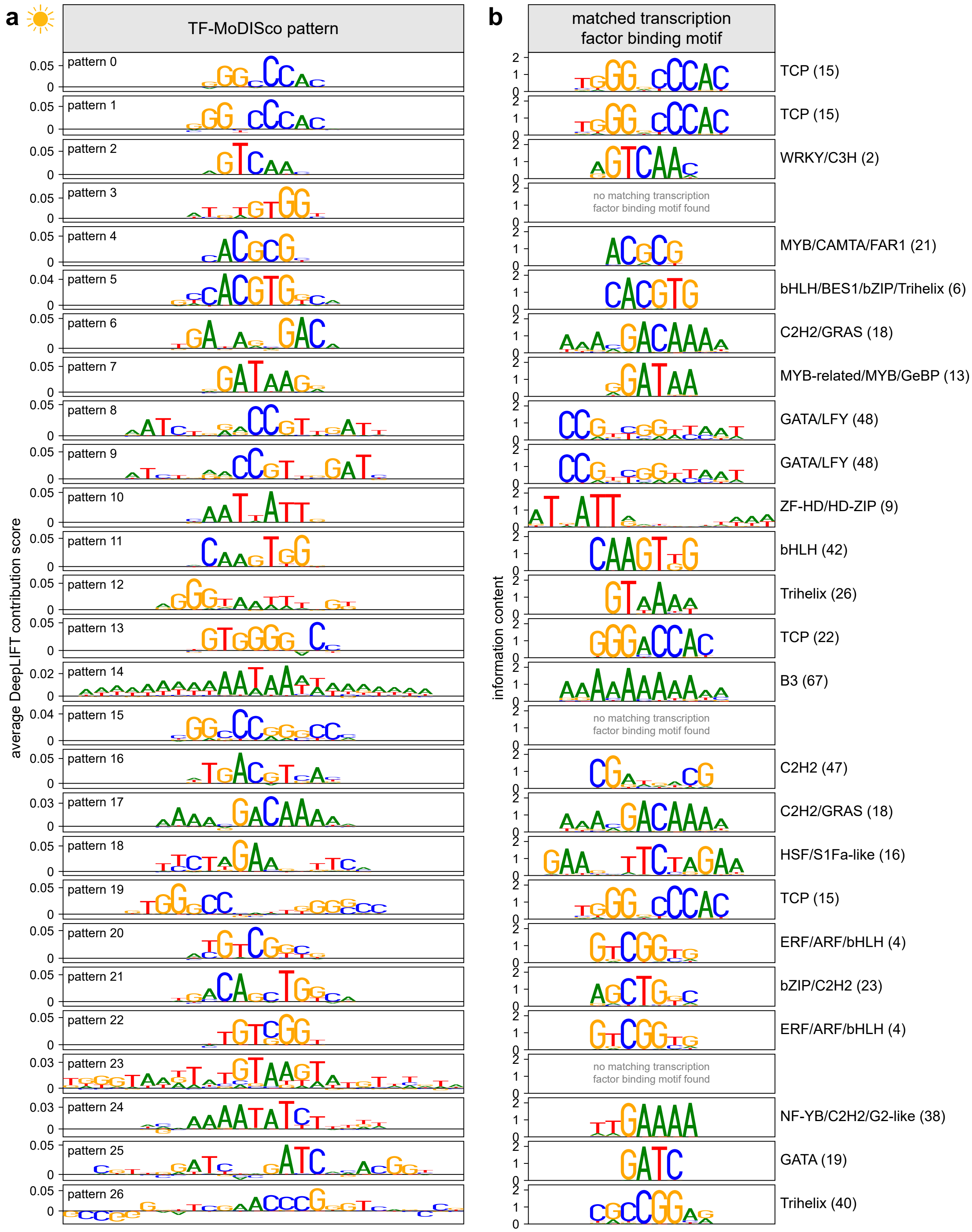
Supplementary Fig. 9 | Removing transcription factor binding sites affects enhancer strength. Relative enhancer strength of sequences with mutations destroying a single strong (*p* value ≤ 0.0001) binding site for transcription factors of the indicated family. For each sequence, the enhancer strength was normalized to the strength of the corresponding wild-type sequence. Box plots represent the median (center line), upper and lower quartiles (box limits), 1.5× interquartile range (whiskers), and outliers (points) for all corresponding sequences. Numbers at the bottom of the plots indicate the number of samples in each group. Significant differences from a null distribution were determined using one-sample Wilcoxon rank-sum tests. Box plots for significant groups (Bonferroni-adjusted *p* value ≤ 0.05) are shown in green. Non-significant groups are represented by gray box plots. Exact *p* values are listed in Supplementary Data 4.



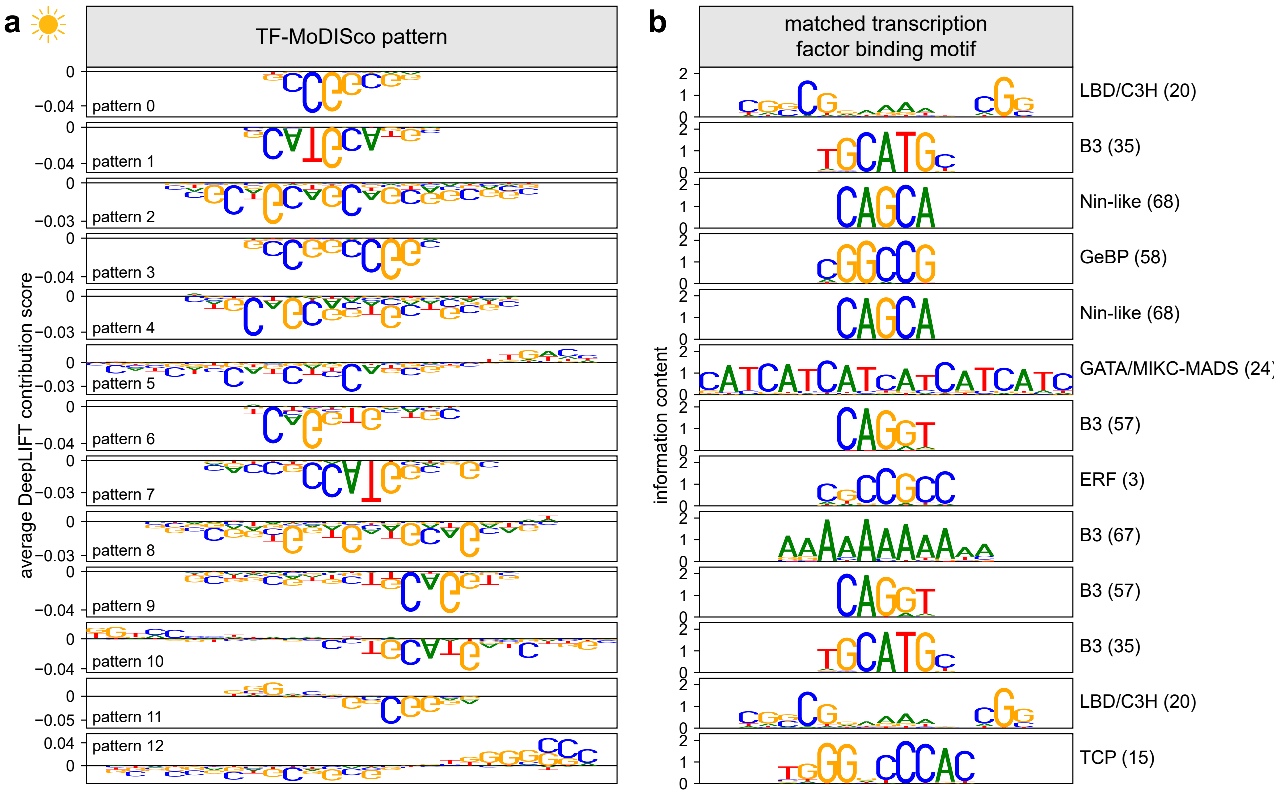
Supplementary Fig. 10 | Sequence context has a stronger impact on enhancer strength than position of a transcription factor binding site. A transcription factor binding site (TFBS) was inserted into 50 random sequences in one of three different positions, and the effect size of each binding site was determined as the fold-change in enhancer strength between the sequence with and without this binding site. For each binding site, the standard deviation of its effects sizes was calculated across the three positions within each sequence (across positions) or across all random sequences for each position (across sequences). Violin plots are as defined in Supplementary Fig. 4. Significant differences were determined by two-sided Wilcoxon rank-sum tests: *, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$; ns, not significant. Exact p values are listed in Supplementary Data 4.



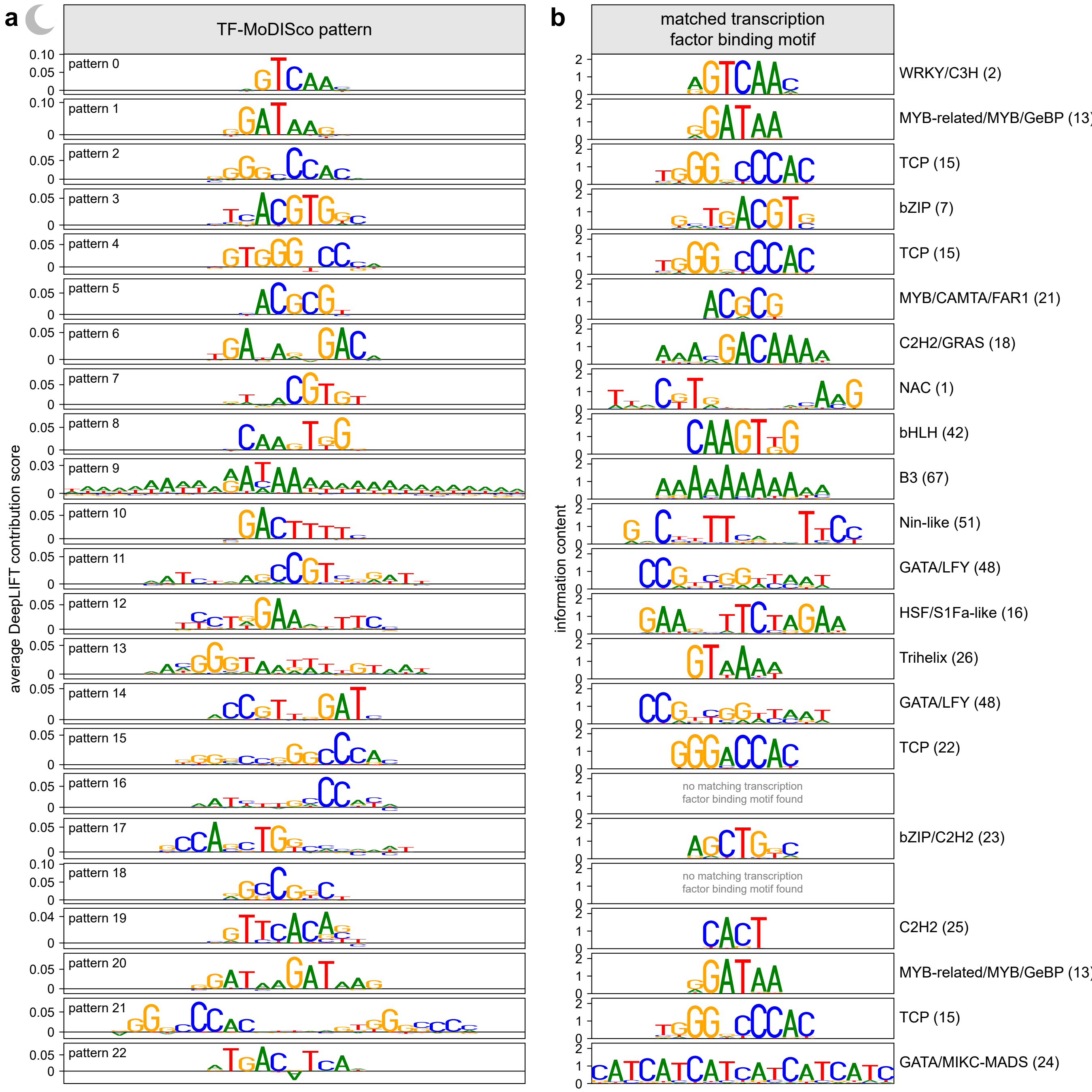
Supplementary Fig. 11 | Linear model coefficients are correlated with the effects of transcription factor binding sites in the ACR library. Correlation between the coefficients of the linear models from Extended Data Fig. 9 and the transcription factor binding site effects in the ACR library (see Fig. 2a). The solid and dashed lines represents a $y = x$ line and a linear regression line, respectively. Pearson's R^2, Spearman's ρ, and the number (n) of samples are indicated.



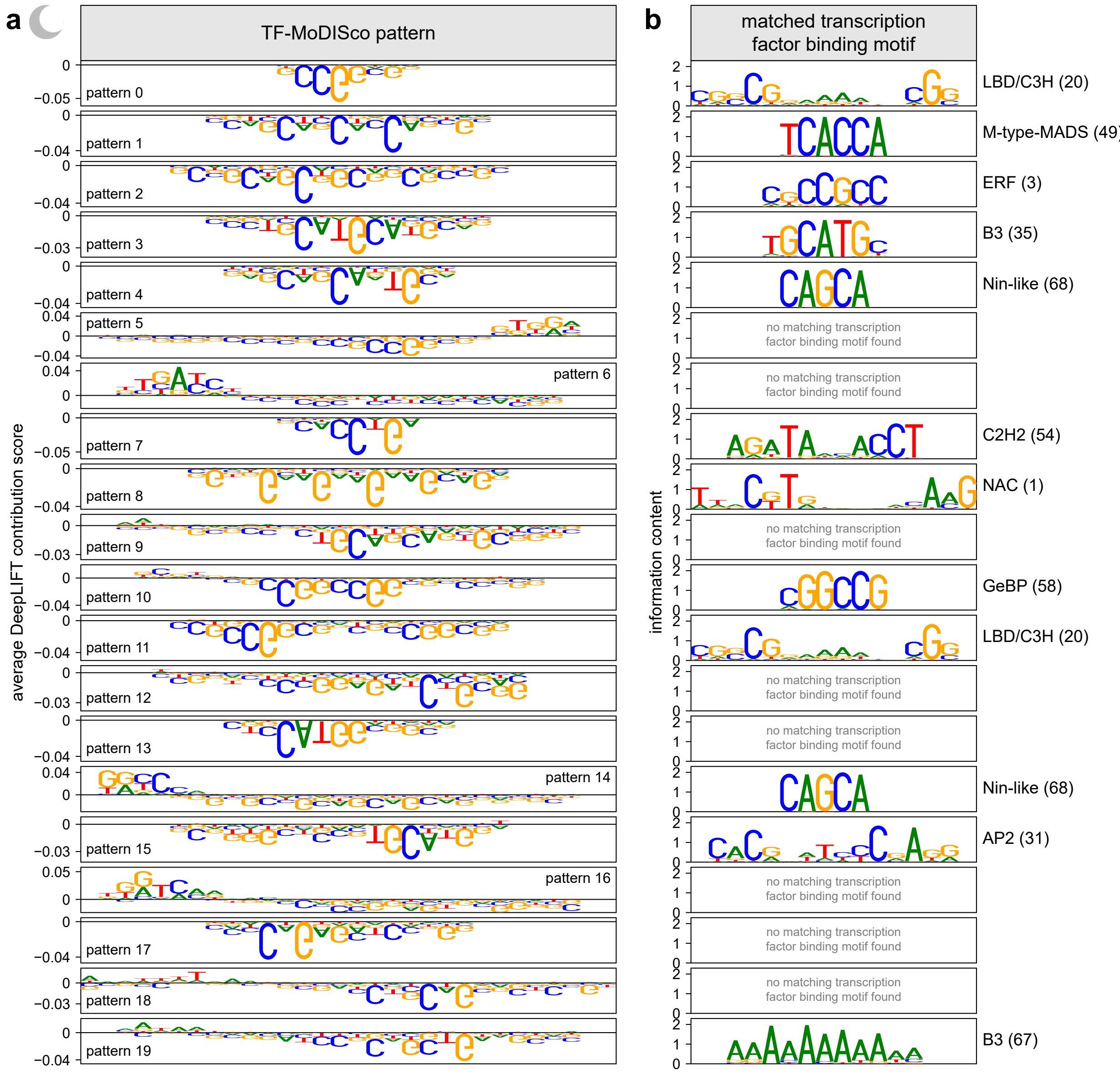
Supplementary Fig. 12 | TF-MoDISco patterns match known transcription factor binding motifs. a, DNA Patterns identified by TF-MoDISco that positively contribute to plantGREP predictions of enhancer strength in tobacco leaves in the light. **b**, Transcription factor binding motifs that match the patterns in (**a**).



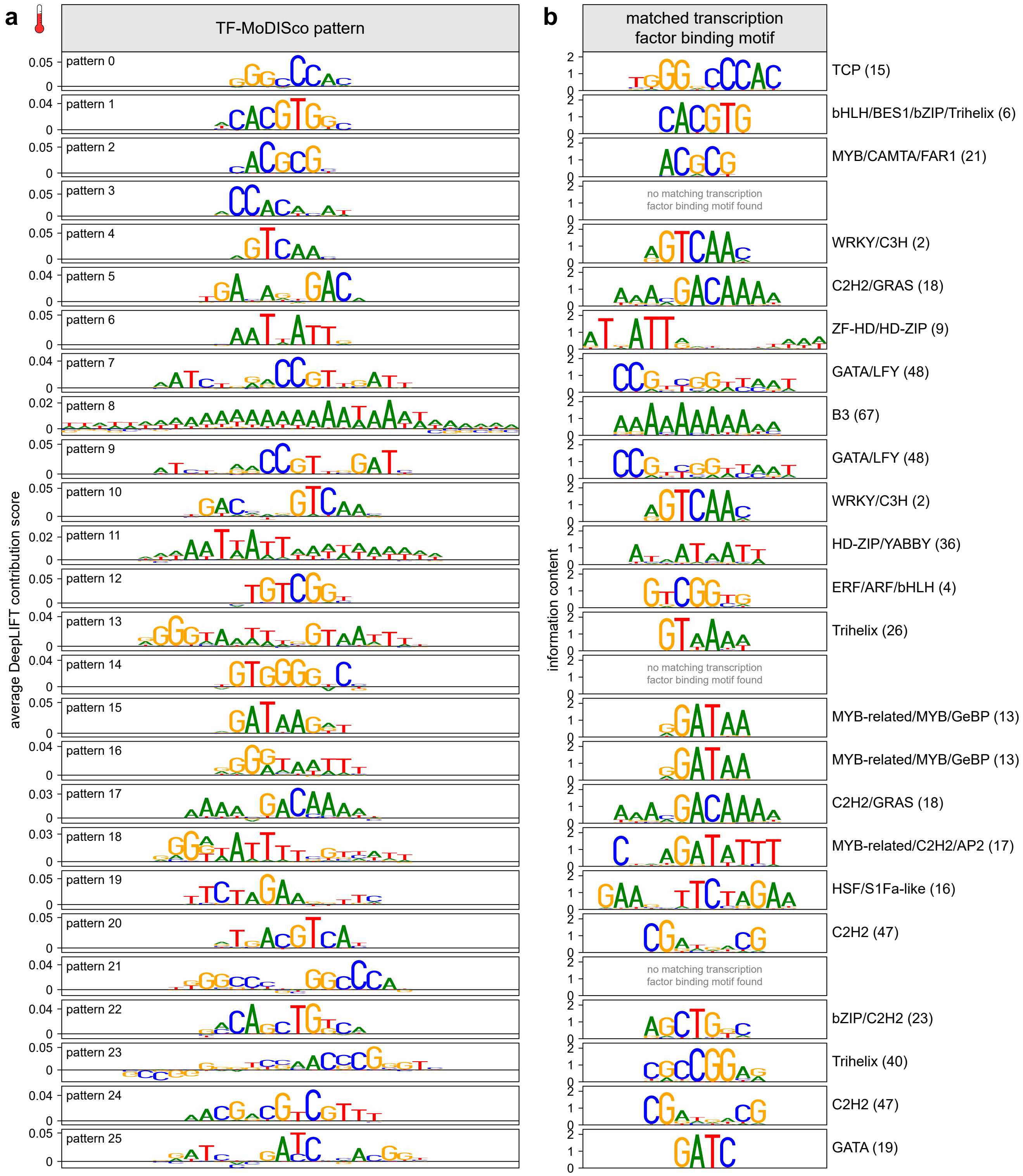
Supplementary Fig. 13 | TF-MoDISco patterns match known transcription factor binding motifs. a, DNA Patterns identified by TF-MoDISco that negatively contribute to plantGREP predictions of enhancer strength in tobacco leaves in the light. **b**, Transcription factor binding motifs that match the patterns in (**a**).



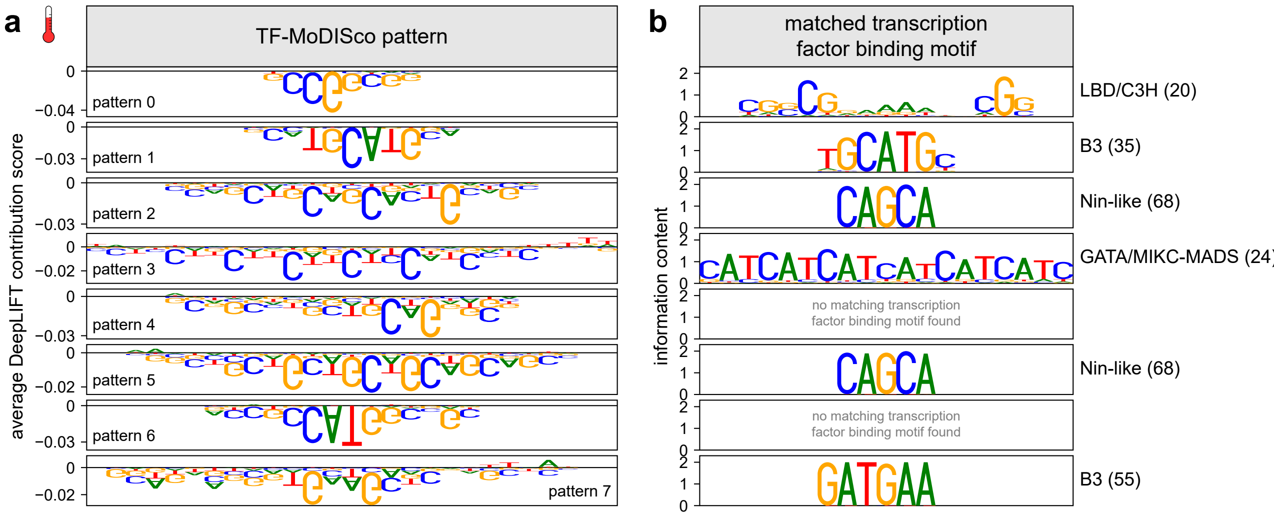
Supplementary Fig. 14 | TF-MoDISco patterns match known transcription factor binding motifs. a, DNA Patterns identified by TF-MoDISco that positively contribute to plantGREP predictions of enhancer strength in tobacco leaves in the dark. **b**, Transcription factor binding motifs that match the patterns in (**a**).



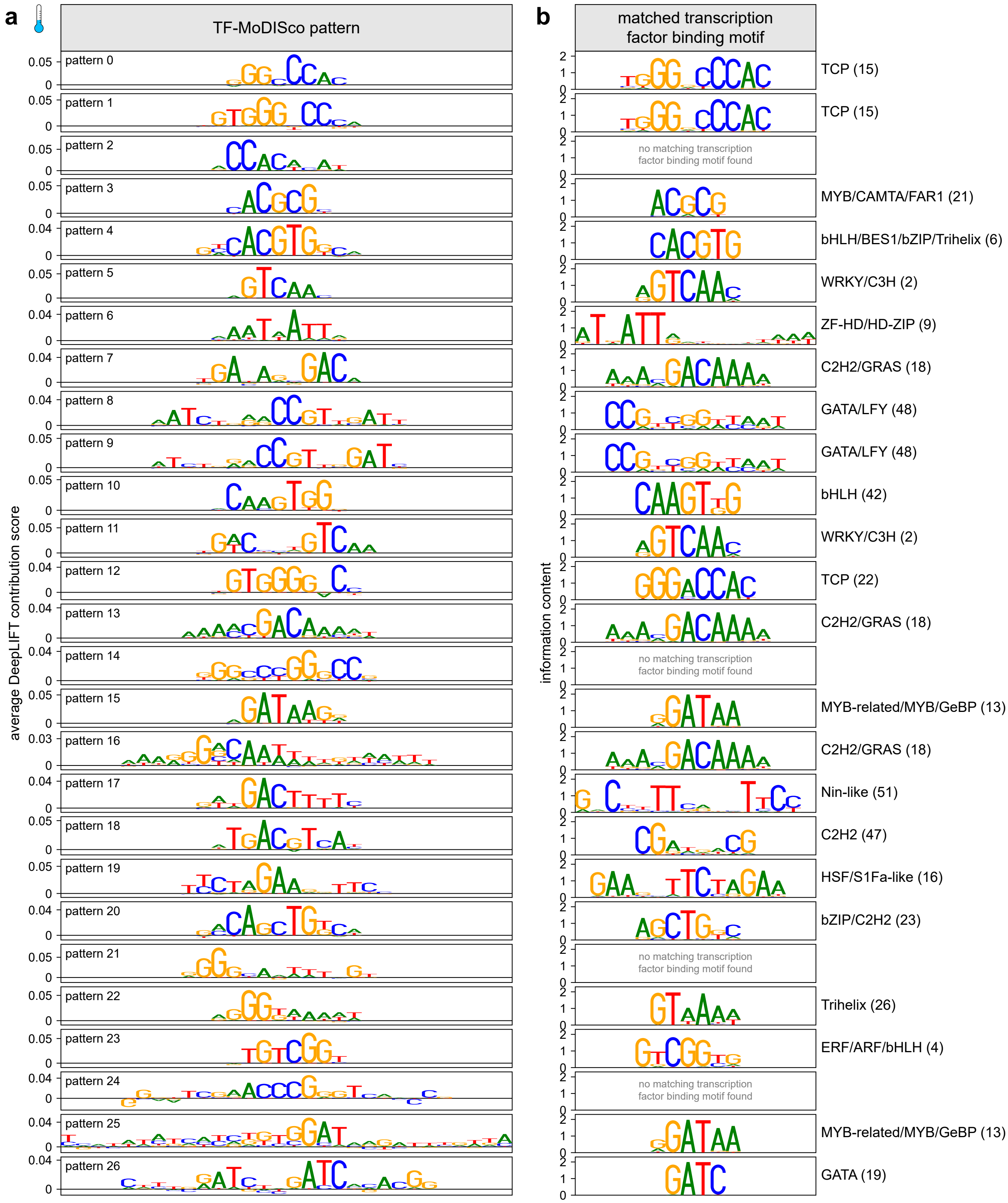
Supplementary Fig. 15 | TF-MoDISco patterns match known transcription factor binding motifs. **a**, DNA Patterns identified by TF-MoDISco that negatively contribute to plantGREP predictions of enhancer strength in tobacco leaves in the dark. **b**, Transcription factor binding motifs that match the patterns in (**a**).



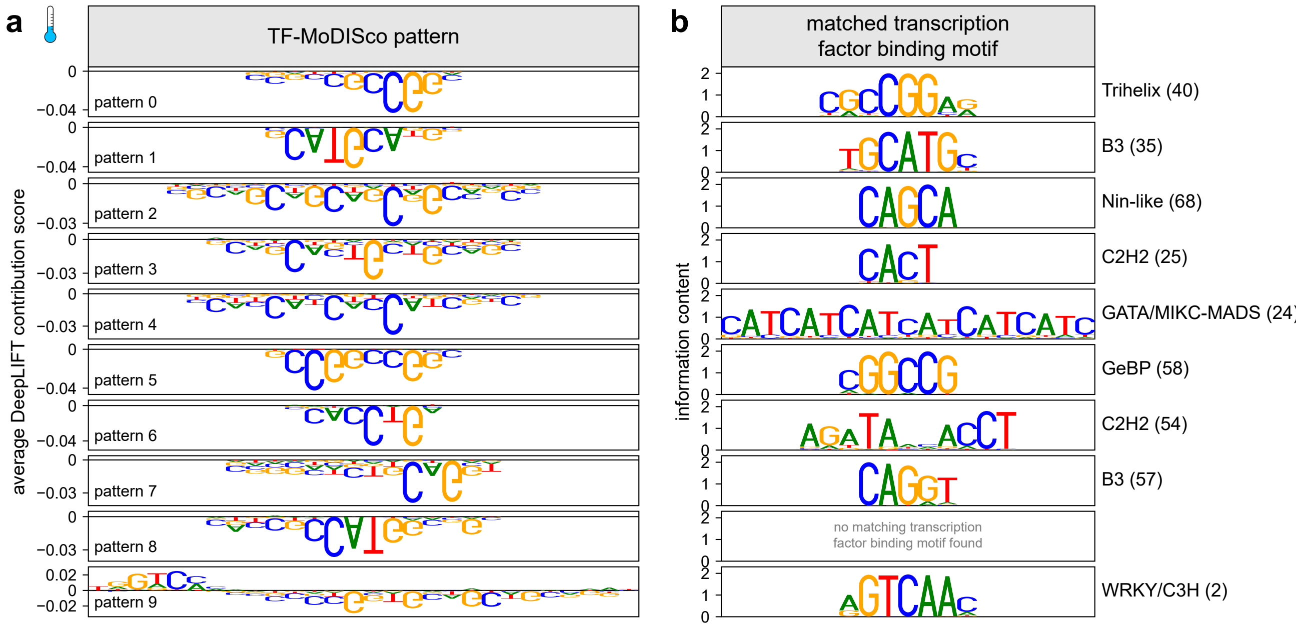
Supplementary Fig. 16 | TF-MoDISco patterns match known transcription factor binding motifs. a, DNA Patterns identified by TF-MoDISco that positively contribute to plantGREP predictions of enhancer strength in tobacco leaves at elevated ambient temperature. **b**, Transcription factor binding motifs that match the patterns in (**a**).



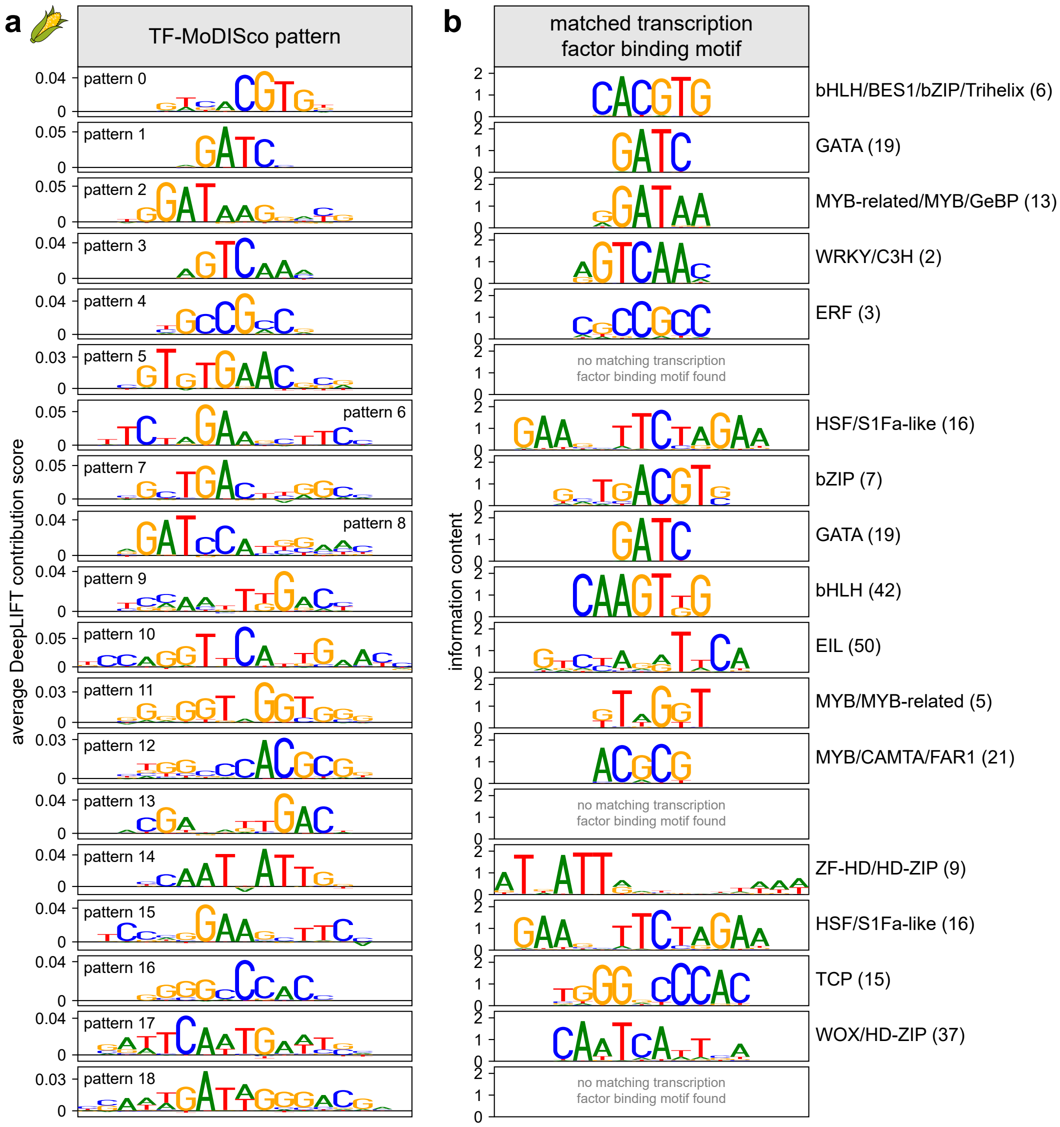
Supplementary Fig. 17 | TF-MoDISco patterns match known transcription factor binding motifs. a, DNA Patterns identified by TF-MoDISco that negatively contribute to plantGREP predictions of enhancer strength in tobacco leaves at elevated ambient temperature. **b**, Transcription factor binding motifs that match the patterns in (**a**).



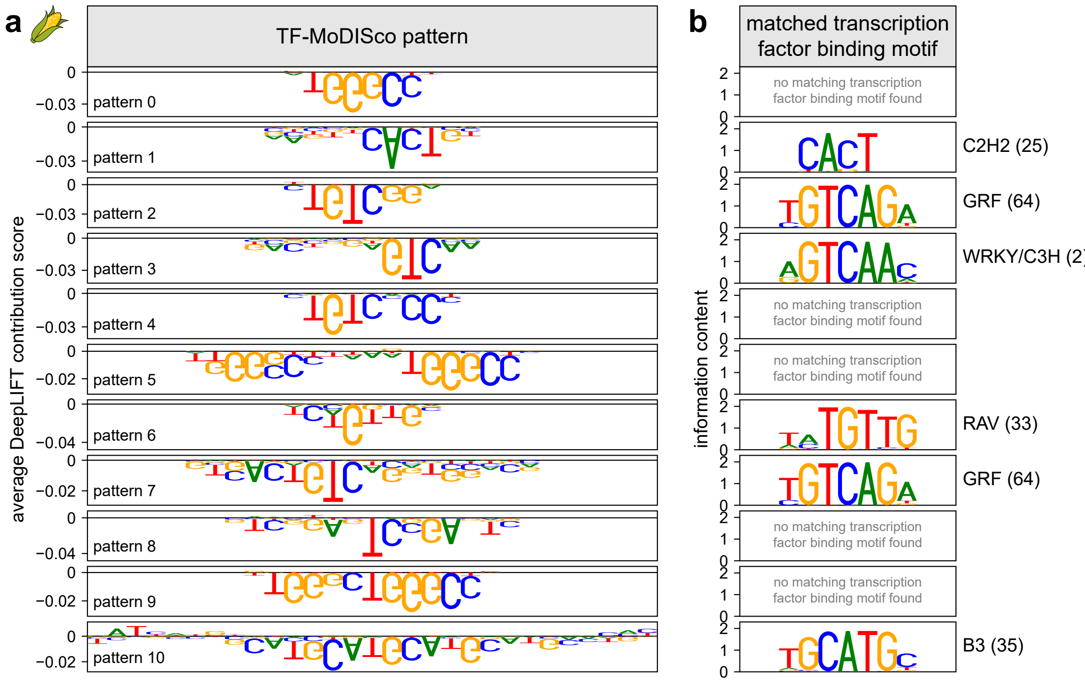
Supplementary Fig. 18 | TF-MoDISco patterns match known transcription factor binding motifs. a, DNA Patterns identified by TF-MoDISco that positively contribute to plantGREP predictions of enhancer strength in tobacco leaves at reduced ambient temperature. **b**, Transcription factor binding motifs that match the patterns in (**a**).



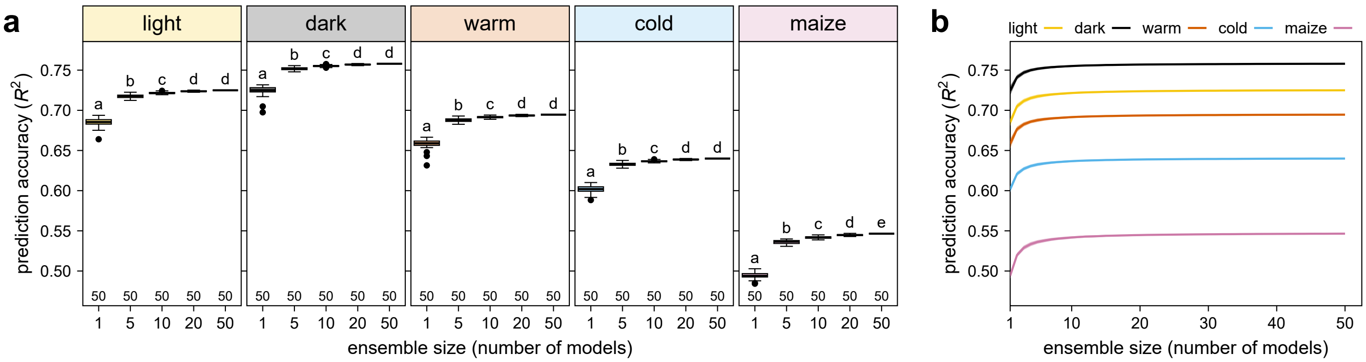
Supplementary Fig. 19 | TF-MoDISco patterns match known transcription factor binding motifs. a, DNA Patterns identified by TF-MoDISco that negatively contribute to plantGREP predictions of enhancer strength in tobacco leaves at reduced ambient temperature. **b**, Transcription factor binding motifs that match the patterns in (**a**).



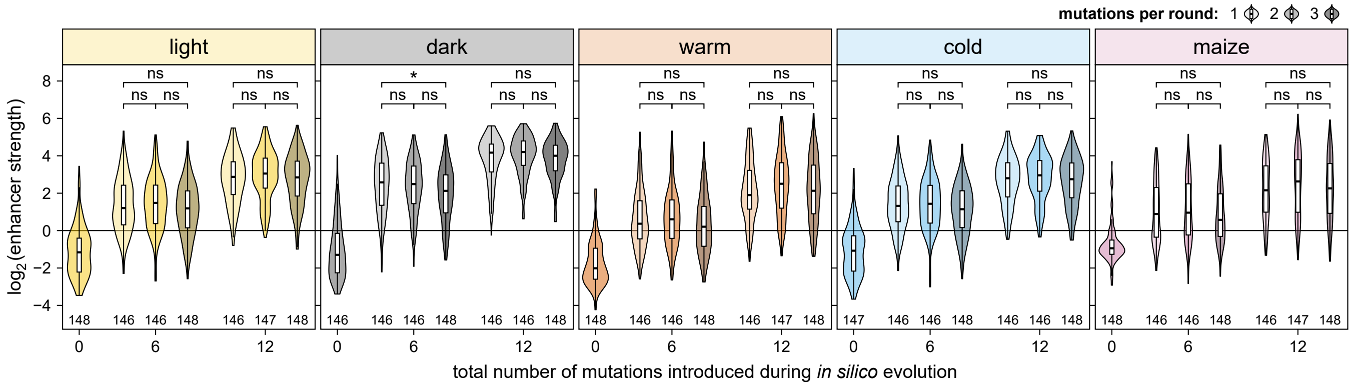
Supplementary Fig. 20 | TF-MoDISco patterns match known transcription factor binding motifs. a, DNA Patterns identified by TF-MoDISco that positively contribute to plantGREP predictions of enhancer strength in maize protoplasts. **b**, Transcription factor binding motifs that match the patterns in (**a**).



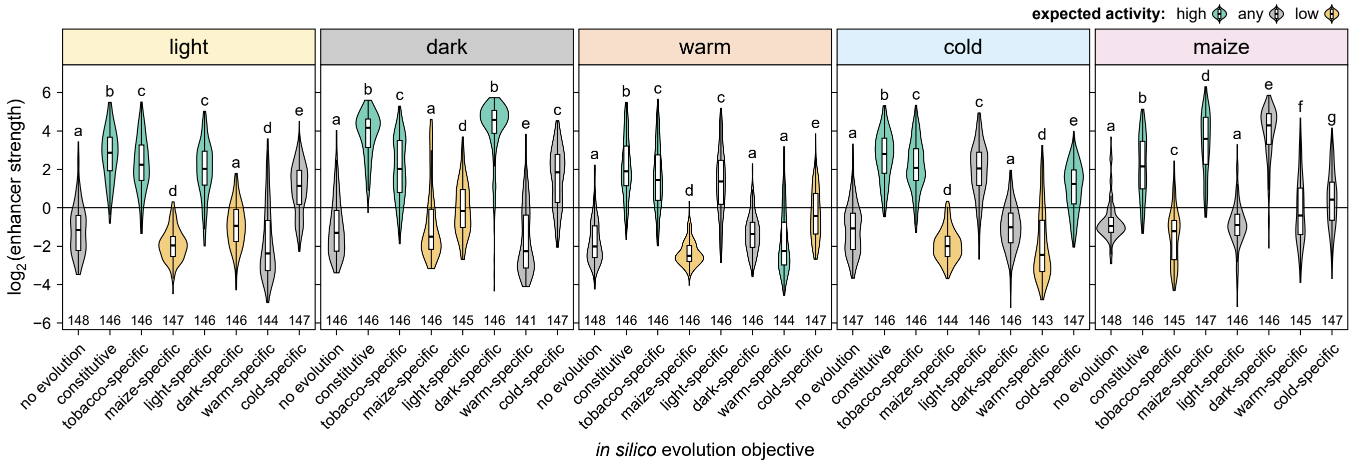
Supplementary Fig. 21 | TF-MoDISco patterns match known transcription factor binding motifs. a, DNA Patterns identified by TF-MoDISco that negatively contribute to plantGREP predictions of enhancer strength in maize protoplasts. **b**, Transcription factor binding motifs that match the patterns in (**a**).



Supplementary Fig. 22 | A model ensemble increases the prediction accuracy of plantGREP. a,**b**, To build a plantGREP model ensemble, 50 models were trained with different, random initializations of the starting model weights. Box plots (as defined in Supplementary Fig. 9) in (**a**) show the prediction accuracy of a plantGREP model ensemble consisting of 1, 5, 10, 20, or 50 randomly sampled models. Lines in (**b**) indicate the mean prediction accuracy and the shaded area indicates the mean ± standard deviation. For each ensemble size, models were sampled independently for a total of 50 times. Letters above the plots in (**a**) indicate significance groups determined by post-hoc Tukey tests performed separately for each condition and species. Exact p values are listed in Supplementary Data 4.



Supplementary Fig. 23 | Increasing the number of mutations per round during *in silico* evolution did not increase performance. *In silico* evolution was performed as in Fig. 4a (*i.e.*, with one mutation per round) or by inserting two or three mutations within an 8-bp window in each round. Sequences before (0 mutations) or after inserting six or twelve mutations during *in silico* evolution were subjected to Plant STARR-seq in the indicated conditions and species. Violin plots are as defined in Supplementary Fig. 4. Significant differences between sequences generated with one, two, or three mutations per round were determined by two-sided Wilcoxon rank-sum tests. The obtained p values were Bonferroni-adjusted separately for each condition and species and are indicated: *, $p \leq 0.05$; **, $p \leq 0.01$; ***, $p \leq 0.001$; ns, not significant. Exact p values are listed in Supplementary Data 4.



Supplementary Fig. 24 | *In silico* evolved enhancers show species- and condition-specific activity. Violin plots (as defined in Supplementary Fig. 4 and colored according to the expected activity in the indicated condition) of the strength of enhancers before (no evolution) or after twelve rounds of *in silico* evolution using plantGREP with different objectives to generate constitutive, species-, or condition-specific enhancers. Letters above the plots indicate significance groups determined by post-hoc Tukey tests performed separately for each condition and species. Exact *p* values are listed in Supplementary Data 4.

Supplementary Table 1 | Composition of the ACR library.

category	*Arabidopsis*	tomato	maize	sorghum	total
ACR sequences	58,068	24,166	52,282	35,314	169,830
non-ACR sequences	1,673	2,164	1,561	1,066	6,464
genes of interest		15,670			15,670
known regulatory elements					36
					192,000

Supplementary Table 2 | Tomato genes of interest with sequences tiled across their upstream regions.

gene ID	description	# sequences
Solyc01g009560	Terminal flower 1	79
Solyc01g009580	Terminal flower 1	76
Solyc01g014100	Unknown Protein	177
Solyc01g066950	Flowering promoting factor-like 1	177
Solyc01g066970	Flowering promoting factor-like 1	83
Solyc01g066980	Flowering promoting factor-like 1	177
Solyc01g080770	Receptor like kinase, RLK	177
Solyc01g087990	MADS-box transcription factor 3	68
Solyc01g092950	MADS-box transcription factor 2	92
Solyc01g093960	MADS box transcription factor	87
Solyc01g098390	GID1-like gibberellin receptor	82
Solyc01g098890	Unknown Protein	28
Solyc01g103530	Receptor like kinase, RLK	177
Solyc02g032180	Nuclear transcription factor Y subunit B-9	177
Solyc02g065290	Dof zinc finger protein 6	177
Solyc02g065730	MADS box transcription factor	59
Solyc02g067550	Unknown Protein	21
Solyc02g069510	Light-dependent short hypocotyls 1	71
Solyc02g071730	MADS-box transcription factor AGAMOUS	114
Solyc02g076820	Light-dependent short hypocotyls 1	177
Solyc02g077390	WUSCHEL-related homeobox 11	137
Solyc02g079020	AP2 domain-containing transcription factor	46
Solyc02g079290	SELF PRUNING 2G	168
Solyc02g081670	Fimbriata	78
Solyc02g082670	WUSCHEL-related homeobox 14	80
Solyc02g083520	BZIP transcription factor	83
Solyc02g083950	WUSCHEL-related homeobox-containing protein 4	154
Solyc02g084630	MADS-box transcription factor	177
Solyc02g087470	Unknown Protein	14
Solyc02g089200	MADS-box transcription factor	80
Solyc02g089210	MADS box transcription factor	114
Solyc02g090220	Dof zinc finger protein	88
Solyc02g091840	Receptor like kinase, RLK	76
Solyc02g092050	AP2-like ethylene-responsive transcription factor At1g16060	24
Solyc02g094460	AP2 domain-containing transcription factor	22
Solyc03g006830	MADS-box transcription factor 2	67
Solyc03g007050	Receptor like kinase, RLK	177
Solyc03g019710	MADS-box transcription factor	90
Solyc03g025960	Unknown Protein	14
Solyc03g026050	Terminal flower 1	142
Solyc03g043770	Receptor like kinase, RLK	177
Solyc03g063100	SELF PRUNING 3D	177
Solyc03g096300	WUSCHEL-related homeobox 5	177
Solyc03g114830	MADS box transcription factor	117
Solyc03g114840	MADS box transcription factor 11	177
Solyc03g115850	NAC domain protein IPR003441	117
Solyc03g117230	Ethylene responsive transcription factor 12	82
Solyc03g118160	FLORICAULA/LEAFY-like protein	54
Solyc03g118770	WUSCHEL-related homeobox-containing protein 4	133
Solyc03g119100	Terminal flower 1	30
Solyc03g123430	AP2-like ethylene-responsive transcription factor At1g16060	26
Solyc04g005320	MADS-box transcription factor	105
Solyc04g009980	Light-dependent short hypocotyls 1	61
Solyc04g015060	Nuclear transcription factor Y subunit B-6	36
Solyc04g040220	NPR1-like protein	177
Solyc04g056640	LRR receptor-like serine/threonine-protein kinase, RLP	158
Solyc04g072570	Receptor like kinase, RLK	92
Solyc04g076280	MPF2-like	71
Solyc04g076680	MADS-box transcription factor-like protein	177
Solyc04g077490	AP2-like ethylene-responsive transcription factor At1g16060	127
Solyc04g078390	F-box protein	177

gene ID	description	# sequences
Solyc04g078650	WUSCHEL-related homeobox-containing protein 4	154
Solyc04g080080	Glycosyltransferase family 77 protein	29
Solyc04g081000	MADS box transcription factor	66
Solyc04g081590	Receptor like kinase, RLK	107
Solyc05g006610	Unknown Protein	177
Solyc05g007650	Unknown Protein	55
Solyc05g056620	Macrocalyx	33
Solyc05g012020	Ripening Inhibitor	177
Solyc05g015720	MADS box transcription factor-like protein	148
Solyc05g015730	MADS-box transcription factor 1	120
Solyc05g015750	Transcription factor MADS-box 2	177
Solyc05g023760	Receptor-like protein kinase 5	53
Solyc05g053630	Unknown Protein	79
Solyc05g053850	SELF PRUNING 5G	117
Solyc05g055020	Light-dependent short hypocotyls 1	55
Solyc05g055660	SELF PRUNING 6A	68
Solyc06g008870	GID1-like gibberellin receptor	95
Solyc06g035570	Transcription factor MADS-box	177
Solyc06g059970	MADS box transcription factor	118
Solyc06g064840	Agamous MADS-box transcription factor	168
Solyc06g069430	MADS box transcription factor	95
Solyc06g069710	NAC domain protein IPR003441	86
Solyc06g072890	WUSCHEL-related homeobox 11	39
Solyc06g074060	Unknown Protein	49
Solyc06g074350	self-pruning	77
Solyc06g076000	WUSCHEL-related homeobox-containing protein 4	62
Solyc06g082210	Light-dependent short hypocotyls 1	115
Solyc06g082520	AP2 domain-containing transcription factor	70
Solyc06g083590	B3 domain-containing transcription factor ABI3	53
Solyc06g083600	B3 domain-containing transcription factor ABI3	53
Solyc06g083860	Light-dependent short hypocotyls 1	64
Solyc07g021170	Os04g0226400 protein	177
Solyc07g053370	Unknown Protein	93
Solyc07g055920	Agamous MADS-box transcription factor	177
Solyc07g062470	Light-dependent short hypocotyls 1	89
Solyc07g062670	Unknown Protein	177
Solyc07g062840	NAC domain protein IPR003441	108
Solyc08g041770	Os04g0226400 protein	177
Solyc08g061560	Receptor like kinase, RLK	177
Solyc08g067230	MADS box transcription factor	26
Solyc08g075340	Glycosyltransferase-like protein	68
Solyc08g078800	GRAS family transcription factor	5
Solyc08g080100	MADS box transcription factor	141
Solyc09g007260	AP2-like ethylene-responsive transcription factor At1g79700	71
Solyc09g009560	SELF PRUNING 9D	66
Solyc09g025280	Light-dependent short hypocotyls 1	54
Solyc09g061410	Unknown Protein	177
Solyc09g074270	Acetyl esterase	177
Solyc09g090180	Light-dependent short hypocotyls 1	177
Solyc09g091810	Unknown Protein	90
Solyc10g007310	Light-dependent short hypocotyls 1	136
Solyc10g008000	Light-dependent short hypocotyls 1	177
Solyc10g017630	MADS-box transcription factor	43
Solyc10g017640	MADS-box transcription factor	177
Solyc10g075030	AP2 domain-containing transcription factor	122
Solyc10g079460	NPR1-like protein	177
Solyc10g079750	NPR1-like protein	177
Solyc11g005120	MADS-box transcription factor 29	50
Solyc11g008640	Flowering locus T	115
Solyc11g008650	Flowering locus T1	27
Solyc11g008660	Flowering locus T	114
Solyc11g010570	MADS box transcription factor	130
Solyc11g010710	AP2-like ethylene-responsive transcription factor At1g16060	177

gene ID	description	# sequences
Solyc11g011260	GAI-like protein 1	177
Solyc11g028020	Agamous MADS-box transcription factor	177
Solyc11g032100	MADS box transcription factor	177
Solyc11g064850	Os04g0226400 protein	149
Solyc11g066120	Unknown Protein	77
Solyc12g038510	MADS box transcription factor 11	177
Solyc11g071380	Unknown Protein	65
Solyc11g072770	WUSCHEL-related homeobox 3B	110
Solyc11g072790	WUSCHEL-related homeobox 3B	106
Solyc12g006340	Auxin response factor 6	146
Solyc12g014260	Light-dependent short hypocotyls 1	69
Solyc12g044760	Os04g0226400 protein	45
Solyc12g056460	MADS box transcription factor	127
Solyc12g087810	Flowering locus C-like MADS-box protein	94
Solyc12g087820	MADS-box transcription factor 13	42
Solyc12g087830	MADS box transcription factor	42
Solyc12g088090	MADS-box transcription factor-like protein	163
Solyc12g099610	Flowering promoting factor-like 1	34

Supplementary Table 3 | Regulatory activity of test sequences.

category[a]	light	dark	warm	cold	maize	at least one condition or assay system	all conditions and assay systems
enhancing sequences	13.6%	19.3%	22.1%	11.1%	4.7%	33.0%	0.9%
inactive sequences	75.5%	69.2%	76.1%	77.9%	84.3%	98.7%	42.9%
repressive sequences	10.9%	11.5%	1.8%	11.0%	11%	26.7%	0.3%

[a] Categories: enhancing sequences, $\log_2$(enhancer strength) > 1; inactive sequences, $\log_2$(enhancer strength) between −2 and 1; repressive sequences, $\log_2$(enhancer strength) < −2

Supplementary Table 4 | Consensus binding motifs for transcription factors of the indicated families. Consensus motifs were obtained from ref. [39]

motif ID	motif logo (forward)	motif logo (reverse-complement)	transcription factor families
1	TACGTA…AAG	CTT…CACG…A	NAC
2	AGTCAAC	GTTGACT	WRKY, C3H
3	CGCCGCC	GGCGGCG	ERF
4	GTCGGTG	CACCGAC	ERF, ARF, bHLH
5	GTAGGT	ACCTAC	MYB, MYB-related
6	CACGTG	CACGTG	bHLH, BES1, bZIP, Trihelix
7	GTGACGTG	CACGTCAC	bZIP
8	AAAAG	CTTTT	Dof, C3H
9	ATATTA…AAA	TTTA…TAATAT	ZF-HD, HD-ZIP
10	CCGTTA	TAACGG	MYB, Trihelix
11	ATATTC	GAATAT	G2-like
12	TCCAAAAAGAAA	TTCCTTTTGGA	MIKC-MADS
13	GGATAA	TTATCC	MYB-related, MYB, GeBP
14	CGTAC	GTACG	SBP, AP2, C2H2, bHLH
15	TGGGCCCAC	GTGGGCCCA	TCP
16	GAATTCTAGAA	TTCTAGAATTC	HSF, S1Fa-like
17	CAGATATTT	AAATATCTG	MYB-related, C2H2, AP2
18	AAAAGACAAAA	TTTTGTCTTT	C2H2, GRAS
19	GATC	GATC	GATA
20	CGCCG…AA…CGG	CCG…TT…CGCG	LBD, C3H
21	ACGCG	CGCGT	MYB, CAMTA, FAR1
22	GGGACCAC	GTGGTCCC	TCP
23	AGCTGC	GCAGCT	bZIP, C2H2
24	CATCATCATCATCATCATC	GATGATGATGATGATGATG	GATA, MIKC-MADS
25	CACT	AGTG	C2H2
26	GTAAAA	TTTTAC	Trihelix
27	TTAGGGT	ACCCTAA	MYB-related, Trihelix
28	TTTGAATTTTAA	TTAAATTCAAA	CPP
29	AGATACG	CGTATCT	ARR-B
30	CATTAATT	AATTAATG	HD-ZIP
31	CACG…TCCA…AG	CCTGG…A…GTG	AP2
32	AATTAGGT	ACCTAAT	MYB
33	TATGTTG	CAACATA	RAV
34	TTTCCCGCCAAAA	TTTTGGCGGGAAA	E2F, DP
35	TGCATGC	GCATGCA	B3
36	ATAATAATT	AATTATTAT	HD-ZIP, YABBY
37	CAATCATTA	TAATGATTG	WOX, HD-ZIP

motif ID	motif logo (forward)	motif logo (reverse-complement)	transcription factor families
38			NF-YB, C2H2, G2-like
39			HD-ZIP, YABBY
40			Trihelix
41			BBR-BPC
42			bHLH
43			E2F, DP
44			Trihelix
45			MIKC-MADS, AP2
46			C2H2
47			C2H2
48			GATA, LFY
49			M-type-MADS
50			EIL
51			Nin-like
52			B3
53			B3
54			C2H2
55			B3
56			B3
57			B3
58			GeBP
59			C2H2
60			SRS
61			C2H2
62			ERF
63			NF-YB
64			GRF
65			Trihelix
66			C2H2
67			B3
68			Nin-like
69			MYB
70			WOX
71			VOZ
72			SRS

Supplementary Table 5 | Oligonucleotides used in this study.

sequence	purpose
GTTCCACTTCTGTATAGGAGCCAATTG	linearize pPSup to replace terminator (fwd primer)
TTACTTGTACAGCTCGTCCATGC	linearize pPSup to replace terminator (rev primer)
TGGACGAGCTGTACAAGTAAGCTCTAGCTAGAGTCGATCGACAAGCTCGAGTTTCTCCATAAT AATGTGTGAGTAGTTCCCAGATAAGGGAATTAGGGTTCCTATAGGGTTTCGCTCATGTGTTGA GCATATAAGAAACCCTTAGTATGTATTTGTATTTGTAAAATACTTCTATCAATAAAATTTCTA ATTCCTAAAACCAAAATCCAGTACTAAAATCCAGATAGGTGTTCCACTTCTGTATAGGAGC	35S terminator for pPSnt
AATTCCACTCCGAGACCACAAGGCGCGCCTAGTGGTCTCCCTGTGCAACCGTCTTCAGTGCCT GCA	replace BsaI cassette for pPSntF (fwd primer)
GGCACTGAAGACGGTTGCACAGGGAGACCACTAGGCGCGCCTTGTGGTCTCGGAGTGG	replace BsaI cassette for pPSntF (rev primer)
AATTCCACAGCGAGACCACAAGGCGCGCCTAGTGGTCTCCGAGTGCAACCGTCTTCAGTGCCT GCA	replace BsaI cassette for pPSntR (fwd primer)
GGCACTGAAGACGGTTGCACTCGGAGACCACTAGGCGCGCCTTGTGGTCTCGCTGTGG	replace BsaI cassette for pPSntR (rev primer)
GCAAGACCCTTCCTCTATATAAGGAAGTTCATTTCATTTGGAGAGGACACGCCCGTCCGAACT CCGAACCCCAGAACAGAGCAAAGCCTCCTCGGCCTCCCTGTCCCCAGCCTTCCCCGATG	template 35S minimal promoter + Zm00001d041672 5′ UTR
GAGAGGAAGACCC-GCAAGACCCTTCCTCTATATAAGGAAGTTCATTTC	amplify 35S minimal promoter + 5′ UTR (fwd primer)
GAGAGGAAGACGGTCAC-NNBNNCNNBNNTNNBNNB-CATCGGGGAAGGCTGGG	amplify 35S minimal promoter + Zm00001d041672 5′ UTR (rev primer for pPSntF)
GAGAGGAAGACGGTCAC-NNBNNTNNBNNCNNBNNB-CATCGGGGAAGGCTGGG	amplify 35S minimal promoter + Zm00001d041672 5′ UTR (rev primer for pPSntR)
GAGAGGAAGACGGTCAC-NNBNNGNNBNNGNNBNNB-CATCGGGGAAGGCTGGG	amplify 35S minimal promoter + Zm00001d041672 5′ UTR (rev primer for no-enhancer control)
CCTTCCTCTATATAAGGAAGTTCATTTCATTTGGAGAGGACACG	linearize pDL to replace BsaI cassette (fwd primer)
GGAATTCGATATCAAGCTTATCGATACCG	linearize pDL to replace BsaI cassette (rev primer)
CGGTATCGATAAGCTTGATATCGAATTCC	amplify BsaI cassette + 35S minimal promoter from pPSnt (fwd primer)
CTCCAAATGAAATGAACTTCCTTATATAGAGGAAG	amplify BsaI cassette + 35S minimal promoter from pPSnt (rev primer)
GAGCGGTCTCCACTC<170-bp enhancer candidate>CTGTAGAGACCGGGC	oligo pool enhancer candidates
GAGCGGTCTCCACTC	amplify oligo pool sequences (fwd primer)
GCCCGGTCTCTACAG	amplify oligo pool sequences (rev primer)
CAATCCGCCCTCACTACAACCGACAGGAAACAGCTATGACCATGATTACGCCAAGCTTGCATG CCTGCAGGTCGACGGTCTCCACTCATTCTAAAGGTCAAAGCTTGATCCGAGCAGAGTATTACT GATTGTATATAATATTCCTAAGTAAATTGAAACTGACATGCATTCAAAGTCAACCTTGCTGCT ACTGCTTATCCGAAGCATCATGGTACTCATCCTTATCTCCCTCTTCTTCAGCCACCTCAGGCA CTGTACTGTCGAGACCTCTAGAGGATCCCCGGGTACCGAGCTCGAATTCACTGGCCGTCGTTT TACAACGCTACTCTGGCGTCGATGAGGGA	gene fragment At-9661
CAATCCGCCCTCACTACAACCGACAGGAAACAGCTATGACCATGATTACGCCAAGCTTGCATG CCTGCAGGTCGACGGTCTCCACTCATAGAAAAATTAGGTAAAGAGTCAGTGTCGTTATGTTAT GGAAGATGTGAATGAAGTTTGACTTCTCATTGTATATGAGTAAAATCTTTTCTTACAAGGGAA GTCCCCAATTGGTCAACATGTGAAAGCACGTGTCATGTTCTTACTTTTGTTTGGGTAATCTTC TAATTCTGTCGAGACCTCTAGAGGATCCCCGGGTACCGAGCTCGAATTCACTGGCCGTCGTTT TACAACGCTACTCTGGCGTCGATGAGGGA	gene fragment Sl-12881
CAATCCGCCCTCACTACAACCGACAGGAAACAGCTATGACCATGATTACGCCAAGCTTGCATG CCTGCAGGTCGACGGTCTCCACTCAATTTAAGAGTAAATGAGAGAAGTTGAATTAGGGCCTTT CGTGGACCAAATTCGTTGCTGCTTTTGGGGTTTAACCCAATTCCTCTTTTTGTATATATCTGT TGTATATCAGTGTATATCGGATGTATACAACCCATTTTTTGTTTTTCCAGACCTTTTGGGGAG CAGAGCTGTCGAGACCTCTAGAGGATCCCCGGGTACCGAGCTCGAATTCACTGGCCGTCGTTT TACAACGCTACTCTGGCGTCGATGAGGGA	gene fragment At-11716
CAATCCGCCCTCACTACAACCGACAGGAAACAGCTATGACCATGATTACGCCAAGCTTGCATG CCTGCAGGTCGACGGTCTCCACTCACTCTTTAGCAACCATCGTCGTCATGCGGCCCATCAACA CACGTGATATCTTATCTTCTCGGGAATTGAGTGCACGCGAGGCGGCCCATTCCCAACAGCTTT TGTTTCGCTGGAGTATCTTGGCGTGCCAGACTGCCAGTGCGTCATGTGATGACCGGTGCAGTG CCCTACTGTCGAGACCTCTAGAGGATCCCCGGGTACCGAGCTCGAATTCACTGGCCGTCGTTT TACAACGCTACTCTGGCGTCGATGAGGGA	gene fragment At-12871
CAATCCGCCCTCACTACAACCGACAGGAAACAGCTATGACCATGATTACGCCAAGCTTGCATG CCTGCAGGTCGACGGTCTCCACTCCATGGCATCATGCTATCATGTGTGGTTGTCGAAGTCTCC ATCCGTTGCGCTTGCGGTACGCCGTTGACCCGGGCAACGCGTGTTCCTTGCCACGCAGGCGAT GATGGATCGGACGGTGCAGACAGCCTCGAGATCCGTGGATCCAGCGGGTAACTAGCCTTCCCC TGTCCCTGTCGAGACCTCTAGAGGATCCCCGGGTACCGAGCTCGAATTCACTGGCCGTCGTTT TACAACGCTACTCTGGCGTCGATGAGGGA	gene fragment At-13524
CAATCCGCCCTCACTACAACCGACAGGAAACAGCTATGACCATGATTACGCCAAGCTTGCATG CCTGCAGGTCGACGGTCTCCACTCATCCAATAGTCATCTTTGTAAAGATTTTTTGTTCGTGTG TGGCGGATAAGCAAAAACAGAATAACATAATCTCTTATCCAACTTATATTGCCACGTGGACAG ATATAGTTGGTGAAGCAGATACTAATCTAATCTGAGCAAAACCAGTTGGGCTATTTGAAAGCC ACAAGCTGTCGAGACCTCTAGAGGATCCCCGGGTACCGAGCTCGAATTCACTGGCCGTCGTTT TACAACGCTACTCTGGCGTCGATGAGGGA	gene fragment At-2020

sequence	purpose
CAATCCGCCCTCACTACAACCGACAGGAAACAGCTATGACCATGATTACGCCAAGCTTGCATG CCTGCAGGTCGACGGTCTCCACTCAATAGACATGGACATTAGGATAGAGAAATTGGATCAAAT TTTCAACTAATTCTCATCCTCAATAGAAGTGGGACCCTTTGAAATCCATTTGACTACAACTTT ATCCAATCAAAAACAAGTAATAAAGATGATGTTTATAATTTTAGCCAATTAAAATACAGATAA GGCCTCTGTCGAGACCTCTAGAGGATCCCCGGGTACCGAGCTCGAATTCACTGGCCGTCGTTT TACAACGCTACTCTGGCGTCGATGAGGGA	gene fragment At-1
CAATCCGCCCTCACTACAACCGACAGGAAACAGCTATGACCATGATTACGCCAAGCTTGCATG CCTGCAGGTCGACGGTCTCCACTCTAAGGTTCCATAAGATCCACAAGAAAAGGATAAAACGGT AAGGAGTGTGCAATGTGCACCACTTAAAATCTATTGCTATTCCAGGCTTAGGCCCATTCTCGG ACCCAATTTCAGTTTTGTTTGGAAGTGTTTCAACTTTTTCTTTTTTTATGTTAACCGATGTGA TTCAACTGTCGAGACCTCTAGAGGATCCCCGGGTACCGAGCTCGAATTCACTGGCCGTCGTTT TACAACGCTACTCTGGCGTCGATGAGGGA	gene fragment Sl-774
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GAACTTGTGGCCGTTTACG	reverse transcription primer for barcode sequencing
AATGATACGGCGACCACCGAGATCTACAC<8-bpindex2>CCCTCGAGGTCGACGGTATC	amplify pPSnt constructs for subassembly (fwd primer)
AATGATACGGCGACCACCGAGATCTACAC<8-bpindex2>CCTCGGCCTCCCTGTCC	amplify pPSnt constructs or reporter cDNA for barcode sequencing (fwd primer)
CAAGCAGAAGACGGCATACGAGAT<8-bpindex1>CACCCCGGTGAACAGCTCC	amplify pPSnt constructs or reporter cDNA for subassembly or barcode sequencing (rev primer)
CTCGAGGTCGACGGTATCGATAAGCTTGATATCGAATTCCAC	NGS read 1 primer for subassembly
CTCCAAATGAAATGAACTTCCTTATATAGAGGAAGGGTCTTGCAC	NGS read 2 primer for subassembly
CCTCGGCCTCCCTGTCCCCAGCCTTCCCCGATG	NGS index 1 primer for subassembly and NGS read 1 primer for barcode sequencing
CACCCCGGTGAACAGCTCCTCGCCCTTGCTCAC	NGS index 2 primer for subassembly and NGS read 2 primer for barcode sequencing
GTGAGCAAGGGCGAGGAGCTGTTCACCGGGGTG	NGS index 1 primer for barcode sequencing
CATCGGGGAAGGCTGGGGACAGGGAGGCCGAGG	NGS index 2 primer for barcode sequencing